National University of Computer and Emerging Sciences, Lahore

AI Automation for Sales and Customer Care

Muhammad Ammar Ali Khan 21L-5659 BS(DS)

Muhammad Hamza Ahmed 21L-6292 BS(DS)

Zain Al Abidin 21L-6260 BS(DS)

Supervisor: Seemab Ayub

Final Year Project

March 19, 2025

Anti-Plagiarism Declaration

This is to declare that the above publication was produced under the:

Title: AI Automation for Sales and Customer Care

is the sole contribution of the author(s), and no part hereof has been reproduced as it is the basis (cut and paste) that can be considered Plagiarism. All referenced parts have been used to argue the idea and cited properly. I/We will be responsible and liable for any consequence if a violation of this declaration is determined.

Date: 8th December 2024

Name: Muhammad Ammar Ali Khan

Signature: ....

Name: Muhammad Hamza Ahmed

Signature: ....

Name: Zain Al Abidin

Signature: ....

Author's Declaration

This states Authors' declaration that the work presented in the report is their own, and has not been submitted/presented previously to any other institution or organization.

Abstract

The AI Automation for Sales and Customer Care project leverages artificial intelligence to streamline sales and customer support by automating lead scoring, customer interactions, and the overall sales process. Integrated within CRM systems, the platform includes AI-driven client calls, chatbots, order processing, sentiment analysis, and real-time report generation. It analyzes customer input to categorize leads, provide relevant offers, and escalate complex issues to human agents only when necessary. Designed for scalability and security, the platform reduces repetitive tasks and enhances the customer experience—helping businesses boost conversion rates and overall performance by streamlining workflows and making customer interactions smoother.

Executive Summary

This project brings up one of the major issues facing businesses; that is, the inefficiencies regarding how sales and service to the customers are managed because fragmentation exists in the systems. These automation platforms with AI shall contribute to the development of AI-controlled automation platforms. They are formulated to ease those tasks which help reduce manual loads and increase efficiency levels. It avails several features such as auto-management of calls, automated customer support enabled through AI-powered chatbots, integration with the system in terms of CRM, and real-time reporting for business decision-makers. All these features will enhance the handling of customers and conversion rates.

The problem that our project solves is the lack of a comprehensive, automated sales and customer care system. Current tools leave a great deal of hard work as it is - one can see this in dealing with customer calls and updating data. Our platform is going to automate everything-from initial client contact to real-time reporting and feedback analysis that enables businesses to cut costs and improve their performance. The project would aim at integration and smooth development of CRM deployment of AI-based customer interactions, which would also ensure automation of real-time reporting and analysis. Ensuring AI for the automation of calls by a customer, response through support response, and further analysis regarding the customer's response would help increase overall satisfaction and make the sales process simpler.

The system comprises several advanced functionalities: AI-Driven Sales Calls: Automation of customer interaction through phone calls using advanced AI techniques to pitch products and services into an individual's customer profile. It should also take customer inquiries, which it should escalate appropriately to humans; the availability of service should be available at all times 24/7. CRM Data Integration-this is how customer data are integrated into the CRM to personally interact with customers at their level. Automated Reporting will also enable companies to personalize real-time sales performance and customer feedback reports so they can make effective decisions.

Sentiment Analysis: The AI process will analyze what consumers are saying, look for sentiment trends, and take action to improve the business.

Technical Solution This system uses the microservices architecture to provide guarantees of scalability and also multiple customer contacts through any number of communication channels. Major subsystems include:

AI Call Subsystem. It also does automated telephone calls which have balanced flows in real-time feedbacks. Scheduler Subsystem: controls the scheduling time gap between interactions of customers and optimizes resource allocation concerning priority and workload. Customer Selection Subsystem:

Algorithms based on CRM databases to identify and rank the clients to follow up. The website is built using the most advanced technologies in web and AI, with frontend frameworks such as Next.js, Backend ones like FastAPI, and ChromaDB managing the AI-driven elements. The large language models for voice AI and chat developed are LLaMA and Phi.

Business Opportunity and Impact This fills up an empty niche in the marketplace regarding the full automation of sales and customer care processes engaged. Operations will be improved with fewer human hands in repetitive tasks and a heightened increase in the satisfaction of clients through relevant and timely interactions. The ability to make minute detail in real-time reports can help the business to respond immediately based on data analysis and thus enable enhancement of productivity and profitability. Other challenges include integration with all the CRM platforms and security while the data is kept stored. Briefly, it is an AI automation for both sales and customer care that is an integrated solution for revolutionizing how technology approaches the performance of handling business. The whole system basically streamlines all key customer-facing functions into optimized business workflows as well as better decision-making patterns to create a better customer experience and further business advantage.

Table of Contents

List of Figures **x**

List of Tables **xi**

1 Introduction **1**

1.1 Purpose of this Document 1

1.2 Intended Audience 1

1.3 Definitions, Acronyms, and Abbreviations 2

1.4 Conclusion 2

2 Project Vision **3**

2.1 Problem Domain Overview 3

2.2 Problem Statement 3

2.3 Problem Elaboration 3

2.4 Goals and Objectives 3

2.5 Project Scope 4

2.6 Sustainable Development Goal (SDG) 5

2.7 Constraints 5

2.8 Business Opportunity 6

2.9 Stakeholders Description/User Characteristics 6

2.9.1 Stakeholders Summary 6

2.9.2 Key High-Level Goals and Problems of Stakeholders 7

3 Literature Review / Related Work **8**

3.1 Definitions, Acronyms, and Abbreviations 8

3.1.1 Definitions 8

3.1.2 Acronyms 8

3.1.3 Abbreviations 8

3.2 Detailed Related Work 9

3.2.1 Salesforce Einstein . . . 9
3.2.2 HubSpot . . . 9
3.2.3 Zendesk . . . 10
3.2.4 BLAND.ai . . . 10
3.2.5 ADA.ai . . . 11
3.2.6 Yellow AI . . . 11
3.2.7 Outreach.io . . . 12
3.3 Related Work Summary Table . . . 12
3.4 Detailed Literature Review . . . 13
3.4.1 DialogBench: Evaluating LLMs as Human-like Dialogue Systems . . . 14
3.4.2 AI Coach Assist: An Automated Approach for Call Recommendation in Contact Centers for Agent Coaching . . . 14
3.4.3 Chat GPT Integrated with Voice Assistant as Learning Oral Chat-based Constructive Communication to Improve Communicative Competence for EFL Learners . . . 15
3.4.4 Real-Time Speech Interruption Analysis: FROM CLOUD TO CLIENT DEPLOYMENT . . . 16
3.4.5 SmartBook: AI-Assisted Situation Report Generation for Intelligence Analysts . 17
3.5 Literature Review Summary Table . . . 17
3.6 Conclusion . . . 18

4 Software Requirement Specifications 19
4.1 List of Features . . . 19
4.2 Functional Requirements . . . 19
4.2.1 User Management . . . 19
4.2.2 CRM Data Integration and Lead Scoring . . . 19
4.2.3 AI-Driven Phone Calls and Product Pitching . . . 20
4.2.4 Customer Support Chatbot . . . 20
4.2.5 Order Placement During Calls . . . 20
4.2.6 Call Feedback and Sentiment Analysis . . . 20
4.2.7 Automated Report Generation . . . 20
4.2.8 Logout . . . 20
4.3 Quality Attributes . . . 21
4.4 Non-Functional Requirements . . . 21
4.4.1 Performance . . . 21

4.4.2 Reliability 21

4.4.3 Scalability 21

4.5 Assumptions 21

4.6 Use Cases 22

4.6.1 Use Case 1: Sign Up 22

4.6.2 Use Case 2: Login 23

4.6.3 Use Case 3: Profile View 23

4.6.4 Use Case 4: Profile Update 24

4.6.5 Use Case 5: Generate Reports 24

4.6.6 Use Case 6: CRM Integration 25

4.6.7 Use Case 7: Chatbot Interaction 26

4.6.8 Use Case 8: Logout 26

4.7 Hardware and Software Requirements 27

4.7.1 Hardware Requirements 27

4.7.2 Software Requirements 27

4.8 Graphical User Interface 27

4.9 ER Diagram 31

4.10 Data Dictionary 31

4.11 Risk Analysis 33

5 High-Level and Low-Level Design **34**

5.1 System Overview 34

5.1.1 Core Functionality 34

5.1.2 System Architecture 35

5.2 Design Considerations 35

5.2.1 Assumptions and Dependencies 35

5.2.2 General Constraints 36

5.2.3 Goals and Guidelines 37

5.2.4 Development Methods 38

5.3 System Architecture 38

5.3.1 Subsystem Architecture 39

5.4 Architectural Strategies 44

5.4.1 Tech Stack 44

5.4.2 User Interface Paradigm 44

5.4.3 Error Detection and Recovery 44

5.4.4 Hosting and Version Control . . . 44
5.4.5 Reuse of Existing Software Components . . . 44
5.4.6 Concurrency and Synchronization . . . 45
5.5 Domain Model/Class Diagram . . . 45
5.6 Policies and Tactics . . . 46
5.6.1 Microservices Communication Protocol . . . 46
5.6.2 Data Persistence Strategy . . . 46
5.6.3 Open Source AI Models . . . 46
5.6.4 Coding Guidelines and Conventions . . . 46
5.6.5 Testing Strategy . . . 46
5.6.6 Monitoring and Logging . . . 47

6 Description of Prototype Developed 48
6.1 Implementation . . . 48
6.1.1 Implementation of Real-Time Calls . . . 48
6.1.2 Sales Agent . . . 48

7 Conclusions 50
7.1 Conclusion . . . 50
7.2 Plans for Future Work . . . 50

List of Figures

2.1 SDG 9: Industry, Innovation, and Infrastructure 5

4.1 Landing Screen 28
4.2 Home Screen 28
4.3 Reporting Screen 29
4.4 CRM Integration Screen 29
4.5 Sales Agent Screen 30
4.6 Login Screen 30
4.7 Database design 31

5.1 High level design of the Project 39
5.2 Architecture Management and Analysis 40
5.3 AI calls workflow Architecture 41
5.4 Architecture of Client Selection Subsystem 42
5.5 AI Scheduler Subsystem 43
5.6 Class Diagram 45

List of Tables

3.1 Summary of Relevant AI Platforms Relevant 13
3.2 Feature Comparison of AI Platforms 13
3.3 Summary of Reviewed Literature 17

4.1 Data Dictionary - Part 1 31
4.2 Data Dictionary - Part 2 32
4.3 Data Dictionary - Part 3 33

Chapter 1 Introduction

As the business grows, so does the customer relation and sales queries. It's really challenging for businesses to handle customer complaints with inefficient and delayed response times. This could be really harmful in terms of customer satisfaction and business performance. This project will simplify these issues by designing an AI-driven solution for customer care and sales. Using such advanced AI tools, in particular, using chatbots and voice assistants, the system enables businesses to respond to all types of customer inquiries with high efficiency. This will reduce responses and reaction times; enhance service quality, which subsequently improves customer experience in the end Optimize internal workflows.

1.1 Purpose of this Document

It presents the design, development, and evaluation of our Final Year Project (FYP), which has as its aim the development of an AI-based automation sales and customer care system. The project focuses on using AI technologies, such as chatbots and voice assistants, to help businesses efficiently manage customer interactions and sales inquiries. The main object of this project would be to come up with an answer that increases the quality of the services coming out for the customer, given the decrease in response times and the generation of better service quality, and to streamline internal workflows.

This document outlines the main methods and techniques of the study, which include AI models, RESTful API communication, and the integration of Next.js for frontend and FastAPI for backend. The document describes the architecture of the system, how it was put to the test, and the different types of limitations that emerged during development. The aim is to give as comprehensive an overview as possible about this project and the results provided.

1.2 Intended Audience

The intended audience for this Final Year Project (FYP) report includes the panel of academic evaluators who will be assessing the project. It is also aimed at industry professionals particularly those who wants to automate their tasks using AI
To identify the intended audience, consider the report's purpose, who will be using the report and for what purpose. For example, the intended audience for a FYP report in Computer Science may be a panel of professors who will be evaluating the project. In contrast, the intended audience for a FYP report in Business may be potential investors or clients.

1.3 Definitions, Acronyms, and Abbreviations

List all important definitions, acronyms, and abbreviations used in this document. For example: **FYP**: Final Year Project

AI: Artificial Intelligence

MVP: Minimum Viable Product

UI: User Interface

UX: User Experience

Agile: Agile Development

SCRUM: Scrum Development Framework

REST: Representational State Transfer

API: Application Programming Interface

gRPC: Remote Procedure Call framework used for microservice communication

ORM: Object-Relational Mapping

CRUD: Create, Read, Update, Delete

FastAPI: A modern web framework for building APIs in Python

Next.js: A React framework used for building web applications

Langchain: A framework for developing applications powered by language models

LlamaIndex: A tool for connecting language models to external data sources

Chroma DB: A vector database used for managing embeddings in AI applications

MongoDB: A NoSQL database used for handling unstructured data

Microservices: A software architecture style where services are small, independent, and loosely coupled

1.4 Conclusion

The report portrays a clear overview of the project, showing its objectives and audience and essential definitions. It develops the vision of the project and identifies core problems and objectives, focusing on relevant sustainable development goals and stakeholder characteristics. Additionally, the report contains a literature review related to the same topic and outlines software requirements necessary for the implementation of the system. It also outlines the proposed methodologies and design strategies that will be pivotal in shaping the development process. Overall, the paper is a good foundation to any study concerning the scope and the scope direction of the project.

Chapter 2 Project Vision

In Chapter 2, we explore the vision behind our AI Automation for Sales and Customer Care project. In this chapter we analyze the critical challenges businesses face in managing sales and customer service operations and highlight the inefficiencies of manual processes. By clearly defining the project's objectives, scope, and challenges we aim to demonstrate how this platform will revolutionize and automate operations. Our focus is on automation and real-time reporting, with the goal of optimizing workflows, enhancing customer satisfaction, and providing a superior experience for stakeholders.

2.1 Problem Domain Overview

Many businesses face challenges with disconnected systems that require manual management of sales and customer support processes. This often leads to time-consuming tasks, errors, and overall inefficiencies in their operations.

2.2 Problem Statement

There is no integrated, fully automated system that can handle the sales and customer care procedures. Solutions present CRM integration or even AI possibilities, but mostly, they never really streamline the whole workflow of sales and customer service. The calls and the data are still processed by sales teams which then have to report back to different stakeholders and there is still potential for delays and inefficiencies involved.

2.3 Problem Elaboration

Existing platforms like Salesforce Einstein and HubSpot provide limited automation, focusing mostlyon predictive analytics and data management. Sales representatives still have to manually call potential leads, requiring skilled labor and consuming valuable time. Additionally, there's a lack of real-time reporting to stakeholders on the effectiveness of these calls and the overall customer experience. When sales calls are made manually, it becomes challenging to quantify customer interactions effectively.

2.4 Goals and Objectives

The main objective of this project is to create an AI-powered automation platform that boosts the effectiveness of sales and customer service activities by reducing manual involvement and increasing process automation. The particular aims encompass:

- CRM Integration: Integrate customer relationship management (CRM) systems to effectively manage and utilize customer data.
- Stakeholder Reporting: Develop a module to generate reports on sales performance, return on investment (ROI), and customer feedback, providing insights into areas of improvement.
- AI Chatbot for Customer Support: Build a chatbot capable of handling customer inquiries and requests, reducing the need for human agents.
- AI-Driven Sales Calls: Implement automated order placements and product pitching during AI-managed calls, tailored to customer profiles and preferences.

2.5 Project Scope

The aim of this project is to create and build a web application for automating sales and customer service functions. The app seeks to offer companies an effective and easy-to-use solution for handling customer engagements and sales processes. The web application will consist of these features:

- **Automated Sales and Customer Support** Use AI models to automate customer inquiries, handle order placements, and manage customer interactions.
- **Real-Time Reporting** Generate reports on sales performance, customer interactions, and feedback to provide insights for business decision-making.
- **Integration with Third-Party AI Services** Seamless integration with AI models like Bland AI and Llama 3.2 for enhanced AI-driven interactions.
- **Microservices-Based Architecture** The application will follow a microservices architecture with an API gateway to ensure modular development and ease of scalability.
- **Customer Relationship Management (CRM)** Integrate CRM systems to manage and utilize customer data effectively.
- **User-Friendly Interface** Provide an easy-to-use graphical user interface (GUI)

The project will be developed using a combination of **Next.js** for the front-end, **Django** for the back-end and **a simple ORM** to handle data management. Given that AI is still advancing everyday, we cannot predict with certainty what will work and what might not. Therefore, we will follow a **SCRUM-based development cycle**, which ensures iterative progress and allows us the flexibility to adapt to new requirements as the project evolves.

The project scope does not include the following:

- Developing dedicated mobile applications (iOS or Android).
- Building custom AI models from scratch (focus will be on integrating existing AI services).

This project scope will serve as a guideline throughout the development process, ensuring that all features and deliverables align with the project's vision and objectives.

2.6 Sustainable Development Goal (SDG)

The project aligns with United Nations Sustainable Development Goal 9: Industry, Innovation, and Infrastructure. By automating the whole sales and customer care processes, this project promotes innovation and technological advancement in business operations. It also enhances infrastructure by reducing the need for manual labor and contributing to more sustainable business practices.

Figure 2.1: SDG 9: Industry, Innovation, and Infrastructure

2.7 Constraints

The main constraints for this project include:

- Data Privacy and Security: Ensuring the safe handling of customer data is a critical challenge. Providing secure data protection during AI-driven interactions is essential to the success of the platform.
- Resource Allocation: Developing such an extensive system requires significant computing power and expertise in AI, which could strain financial and technical resources.
- Latency Issues During Calls: Ensuring low latency during AI-managed calls is essential to maintain a smooth conversation flow. Any delays in AI responses could lead to awkward pauses or

interruptions, diminishing the customer experience and potentially leading to frustration or dissatisfaction

- AI Accuracy: Ensuring that AI interactions feel as natural and human-like as possible is crucial. If the AI fails to provide a smooth and empathetic experience, it could harm customer relationships and reduce the effectiveness of sales efforts.

2.8 Business Opportunity

This project offers a significant business opportunity by fulfilling a critical market demand for a fully automated sales and customer service software with complete end-to-end functionality. By utilizing this AI platform, companies can depend less on human employees for repetitive tasks, which will not only reduce operational expenses but also improve customer satisfaction through quicker and more precise responses. Moreover, the platform's real-time reporting enables businesses to make faster data-driven choices, resulting in enhanced efficiency and greater profits.

2.9 Stakeholders Description/User Characteristics

The participants in the system consist of Business Owners/Executives, Sales Managers, and Customer Support Managers.

2.9.1 Stakeholders Summary

The key stakeholders of this AI-driven automation platform are:

- **Business Owners/Executives**: These stakeholders are responsible for strategic decision-making and evaluating the overall performance of the platform. They utilize the platform's real-time reports to assess sales performance, customer satisfaction, and ROI.
- **Sales Managers**: They oversee the AI-driven sales processes, using the platform to track sales performance and adjust sales strategies based on data insights provided by the system.
- **Customer Support Managers**: Focused on customer satisfaction, they monitor the interactions handled by the AI chatbot and ensure that the system delivers smooth and effective support.

2.9.2 Key High-Level Goals and Problems of Stakeholders

2.9.2.1 Cost Reduction

This AI platform promises cost reduction to companies through the automation of sales and customer care service provision. The platform would save on extensive human resource through taking over repetitive jobs like answering customer's questions and placing orders, thus saving much on labor costs

2.9.2.2 Automating Processes

The system automation, in most key elements of sales and customer support, covers order placement, interaction with customers through chat, and also producing real-time reports. The manual intervention is kept to a bare minimum; thus, the workflow can be smooth for both the in-house teams and the customers.

2.9.2.3 Human-Like Customer Interactions

The development of the platform should be such that the AI interacts with the customers as if interacting with the human race. Utilize state-of-the-art AI methodologies to provide each customer with a unique experience that advances the overall level of satisfaction.

2.9.2.4 Real-Time Reporting

The platform's ability to generate real-time reports provides stakeholders with immediate insights into sales performance, customer interactions, and ROI. This helps business owners and managers make timely, data-driven decisions, improving overall productivity.

Chapter 3 Literature Review / Related Work

This literature review highlights the latest advancements in AI-driven automation for sales and customer care, focusing on cutting-edge products, research, and methodologies. Since the AI is rapidly evolving we explore the most recent innovations that directly align with our project.

3.1 Definitions, Acronyms, and Abbreviations

3.1.1 Definitions

AI: Artificial Intelligence

UI: User Interface

UX: User Experience

Microservices: A software architecture style where services independent, and loosely coupled

LangChain: A framework for developing applications powered by language models

LlamaIndex: A tool for connecting language models to external data sources

Chroma DB: A vector database used for managing embeddings in AI applications

Quadrant DB: A vector database used for managing embeddings in AI applications

MongoDB: A NoSQL database used for handling unstructured data

Agents: Autonomous or semi-autonomous systems that perform tasks dynamically based on predefined conditions

Models: Mathematical and statistical algorithms used for making predictions or recommendations based on data

LLMs: Large Language Models, a type of AI model designed to understand and generate human-like language.

3.1.2 Acronyms

FYP: Final Year Project

3.1.3 Abbreviations

REST: Representational State Transfer

API: Application Programming Interface

ORM: Object-Relational Mapping

CRUD: Create, Read, Update, Delete

3.2 Detailed Related Work

In this section, we will discuss the relevant products and literature that relate to our project. Additionally, a summary table will be provided to highlight the key findings of each product and a comparison of our project with each product.

3.2.1 Salesforce Einstein

3.2.1.1 Summary of the research item

Salesforce Einstein [5] is an AI-powered platform designed to enhance various aspects of Salesforce's CRM capabilities. It's key features include predictive analytics, automated data capture, personalized recommendations, and natural language processing.

3.2.1.2 Critical analysis of the research item

Salesforce Einstein provides a complete and comprehensive AI suite. It's strongest point is it's seamless integration with customer data. However it's weaknesses are that it is highly dependent on Salesforce (Parent Application), which limits it's usability for people who are not familiar withe the usage of salesforce, and it also does not allow much customization.

3.2.1.3 Relationship to the proposed research work

While Salesforce Einstein provides a strong foundation for AI-driven CRM automation, its capabilities are limited when it comes to outreach. Our project on the other hand aims to incorporate real-time voice interactions, call feedback analysis, and order placement during calls.

3.2.2 HubSpot

3.2.2.1 Summary of the research item

HubSpot [4] is a comprehensive marketing, sales, and customer service platform that integrates AI for tasks like marketing automation, lead generation, and customer interactions.

3.2.2.2 Critical analysis of the research item

HubSpot offers a comprehensive platform with AI-powered automation and a strong customer-centric focus, making it a valuable tool for businesses as it help cut back costs of hiring sales staff. However, it's downsides are that it only focuses on managing customer interactions through text based communication channels.

3.2.2.3 Relationship to the proposed research work

Hubspot is a complete package for any business but one major feature that could improve its effectiveness by a lot is call based outreach. Calls could potentially improve the effectiveness of the product by a huge amount.

3.2.3 Zendesk

3.2.3.1 Summary of the research item

Zendesk [7] is one of the best customer service platforms that does tasks like ticket routing, automated responses and sentiment analysis.

3.2.3.2 Critical analysis of the research item

Zendesk provides a comprehensive customer service platform that is designed to provide the best experience for the customer. They recently integreated AI into their workflow as well so they could also engage in live conversations with the customers. However, since the platform is solely designed for customer support, they don't provide sales and outreach.

3.2.3.3 Relationship to the proposed research work

While Zendesk offers AI-powered features for customer service, its capabilities still don't cover a wide range of features that could be integrated, for example real-time call interactions. While Zendesk could potentially be integrated for certain functionalities, a more tailored version of it with some added features would be a lot better.

3.2.4 BLAND.ai

3.2.4.1 Summary of the research item

BLAND.ai [2] is an AI platform focused on voice generation. It offers features like creating highly realistic and natural-sounding voices, customization options for accents and tones, and integration with other software for voice-driven interactions.

3.2.4.2 Critical analysis of the research item

Bland AI excels in high-quality voice generation, offering customization options and strong integration capabilities, which makes it the perfect application for automating calls. However, it limitations are the it provides limited control over pronunciation and is expensive.

3.2.4.3 Relationship to the proposed research work

While BLAND.ai excels in voice generation, our proposed project intends on providing a broader AI platform for sales and customer care. BLAND.ai could potentially be integrated for voice generation aspects, but wouldn't offer call feedback analysis, or other functionalities needed for a complete sales and customer care solution.

3.2.5 ADA.ai

3.2.5.1 Summary of the research item

ADA AI [1]is a customer service platform that automates text-based interactions through chatbots, handling routine tasks such as answering FAQs, providing information, and executing basic actions.

3.2.5.2 Critical analysis of the research item

ADA AI's strengths is its efficiency and ability to automate high-volume, routine inquiries via text. However, its limitations include the absence of real-time voice communication and call analysis, which can lead to less dynamic and personalized customer interactions. Overall, it is a well-suited platform for businesses looking to optimize text-based interactions.

3.2.5.3 Relationship to the proposed research work

While ADA.ai offers automated customer interactions, our project focuses on real-time voice interactions, call feedback analysis, and features beyond ADA.ai's current scope.

3.2.6 Yellow AI

3.2.6.1 Summary of the research item

Yellow AI [6]is a customer engagement platform that utilizes chatbots and voicebots to facilitate interactions across various channels. It focuses on enhancing customer experiences by automating responses to the most common inquiries.

3.2.6.2 Critical analysis of the research item

The strengths of Yellow AI include its ability to automate customer interactions and streamline communication, improving efficiency. It's downsides are that it does not provide call feedback mechanisms, upon which sales representatives could improve the sales techniques and tactics.

3.2.6.3 Relationship to the proposed research work

While Yellow AI offers features like multi-channel engagement and analytics, it does not allow order placement during phone calls. By adding the capability of order placement and more interactive voice functionalities, our project will provide a more holistic solution than Yellow AI's current offerings. While Yellow AI provides a full and complete product for it's audience, it still lacks features that could potentially improve it's customer base by a lot.

3.2.7 Outreach.io

3.2.7.1 Summary of the research item

Outreach.io [11] is a sales engagement platform designed to automate tasks such as email outreach, follow-ups, lead scoring, and performance analytics. It supports task management, predictive analytics, and call review for coaching, enhancing the overall sales process and customer experience.

3.2.7.2 Critical analysis of the research item

While Outreach.io improves efficiency by automating repetitive tasks and optimizing lead management. Outreach.io's automation capabilities significantly reduce manual tasks, allowing sales teams to focus more on strategic efforts. It's lack of voice interactions can limit personalization limit sales. Without live conversations, sales teams may miss opportunities for immediate rapport-building and dynamic customer engagement. This absence can potentially result in a more robotic and scripted sales approach and affect not only the sales but also the overall customer experience.

3.2.7.3 Relationship to the proposed research work

While Outreach.io excels in AI-powered sales engagement, our proposed project focuses on real-time voice interactions, call feedback analysis, and functionalities beyond Outreach.io's current scope.

3.3 Related Work Summary Table

Following is the summary table for the above discussed related work section and the comparison table for all the different features provided by the different products highlighting the features they provide and the features that they lack in.

Table 3.1: Summary of Relevant AI Platforms Relevant

Application	Features	Relevance	Limitations
Salesforce Einstein [5]	Predictive analytics, automated data capture, Chatbot support	Enhances CRM capabilities with AI insights	Limited usability for non-Salesforce users; lacks AI-driven phone calls
HubSpot [4]	Marketing automation, lead generation, conversational bots	Comprehensive AI-powered platform for sales	Focuses on text interactions; lacks real-time voice communication
Zendesk [7]	Ticket routing, automated responses, sentiment analysis	Strong platform for customer service automation	Lacks sales features; no real-time voice interactions
BLAND.ai [2]	Voice generation, customization options, integration capabilities	High-quality voice output for various applications	Limited control over pronunciation; potentially high cost
ADA.ai [1]	Text-based automation, knowledge base integration	Streamlines customer support through automation	No real-time voice communication; limited handling of complex inquiries
Yellow AI [6]	Chatbots, voicebots, multi-channel engagement	Automates customer interactions across channels	Lacks order placement during calls; no direct call feedback
Outreach.io [11]	Email automation, lead scoring, performance analytics	Streamlines sales workflows for efficiency	No real-time voice interactions; may limit personalization

Table 3.2: Feature Comparison of AI Platforms

Features	**Our Project**	**Salesforce Einstein [5]**	**HubSpot [4]**	**Outreach .io [11]**	**Zendesk [7]**	**BLAND .ai [2]**	**ADA .ai [1]**	**Yellow AI [6]**
CRM Integration	Yes	Yes	Yes	Yes	No	No	No	Yes
AI-Driven Calls	Yes	No	No	No	No	Yes	No	No
Stakeholder Reporting	Yes	Yes	No	No	No	No	No	Yes
Customer Support Chatbot	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Call Feedback Analysis	Yes	No	No	No	No	No	No	No

3.4 Detailed Literature Review

In this section, we will discuss the relevant research that relates to our project. Additionally, a summary table will be provided to highlight the key findings from each reviewed paper.

3.4.1 DialogBench: Evaluating LLMs as Human-like Dialogue Systems

3.4.1.1 Summary of the research item

The research paper, ”DialogBench: Evaluating LLMs as Human-like Dialogue Systems,” [**?**] introduced a new benchmark designed to assess the human-likeness of large language models (LLMs) in dialogue interactions. Recognizing the importance of human-like qualities in dialogue systems, The authors proposed a comprehensive evaluation framework consisting of 12 dialogue tasks. These tasks covered a wide range of abilities, including emotional perception, information mastery, and commonsense understanding.

3.4.1.2 Critical analysis of the research item

DialogBench offered a thorough and user-focused assessment of the capabilities of LLMs. Nonetheless, it might not encompass every facet of human-like conversation, and evaluating human-likeness is subjective. Moreover, creating evaluation instances with GPT-4 might lead to biases. Even with these constraints, DialogBench remains a useful resource for researchers and developers focused on dialogue systems.

3.4.1.3 Relationship to the proposed research work

The ”DialogBench” benchmark may assist our project by providing an organized method to assess the human-like features of our AI interactions, guaranteeing that our live customer engagement seems more organic and instinctive. Furthermore, the benchmark might provide us with insights that we can potentially leverage to enhance our conversational flow and create a more human-like experience for customers.

3.4.2 AI Coach Assist: An Automated Approach for Call Recommendation in Contact Centers for Agent Coaching

3.4.2.1 Summary of the research item

This study [**?**] investigates the use of Artificial Intelligence (AI) to improve coaching for contact center agents. The suggested system, AI Coach Assist, utilizes a pre-trained language model to pinpoint calls that are best suited for coaching activities. The model underwent training and assessment using a large dataset from a real-world contact center.

3.4.2.2 Critical analysis of the research item

Although AI Coach Assist demonstrates potential in enhancing contact center agent training, it does have certain limitations. To begin with, the system is not tailored for every type of call or language, and in addition, it demands substantial computing resources to function properly. This may pose a difficulty for organizations that have restricted resources. In spite of these limitations, AI Coach Assist serves as a useful tool for enhancing training for contact center agents.

3.4.2.3 Relationship to the proposed research work

This research paper about AI Coach Assist is highly pertinent to our project concentrated on AI-powered sales automation and customer service. It features a pre-trained language model designed to pinpoint the calls best suited for coaching, and incorporating its AI functionalities into our project could offer significant insights and assistance for sales calls. It also discusses how to maintain performance while remaining resource efficient.

3.4.3 Chat GPT Integrated with Voice Assistant as Learning Oral Chat-based Constructive Communication to Improve Communicative Competence for EFL Learners

3.4.3.1 Summary of the research

This study examines the efficacy of utilizing Chat GPT alongside voice assistants for instructing English as a foreign language (EFL) [10]. The authors of this paper contend that this approach provides multiple benefits compared to conventional teaching methods, as it enables students to engage in speaking practice in real time and obtain instant feedback. The results indicate that Chat GPT may serve as a useful resource for enhancing speaking, listening, and writing abilities, while also offering a customized learning experience.

3.4.3.2 Critical analysis of the research

The study emphasizes the promise of Chat GPT alongside voice assistants as a supportive resource for EFL students. It provides advantages such as ease of use, tailored education, and instant responses. Nonetheless, the drawbacks are that Chat GPT cannot completely take the place of human interaction, and additional research involving larger populations is needed to enhance the generalizability of the results.

3.4.3.3 Relationship to the proposed research work

This study is very pertinent to our project since it tackles issues related to Speech-to-Text (STT) systems, particularly for users with different accents. Through the use of Chat GPT voice assistants, we can adjust to various speaking styles and accents. Incorporating these features into our project may address STT challenges, guaranteeing clearer and more efficient communication in sales and customer service conversations, including those with customers who possess various accents.

3.4.4 Real-Time Speech Interruption Analysis: FROM CLOUD TO CLIENT DEPLOYMENT

3.4.4.1 Summary of the research item

This research [8] addressed the issue of detecting interruptions in speech in real-time during remote meetings. The creators introduced a model called WavLM_SI, designed to detect occurrences when one participant tries to talk over another. This is especially useful in scenarios with several individuals conversing, where delays may lead to them interrupting one another. The research emphasizes the successful enhancement of the True Positive Rate (TPR) for detecting failed speech interruptions by utilizing a larger dataset for training and model fine-tuning. They minimized the model's size and complexity to make it better suited for deployment on client devices.

3.4.4.2 Critical analysis of the research item

The research offers a valuable solution for enhancing communication in remote meetings by detecting failed speech interruptions. The improvements in accuracy and the model's reduced size demonstrate a step forward in a practical application. However, the is no mention of how the model performs in an environment with background noise.

3.4.4.3 Relationship to the proposed research work

This research is highly relevant to our project as it focuses on real-time speech detection, which is crucial for effective communication in customer service and sales calls. In real-life applications, it is very common for people to speak over each other and our aim is that our project handles such scenarios effectively. Integrating similar technology could enhance the quality of AI-driven calls and ensuring a more human-like conversation for the customers

3.4.5 SmartBook: AI-Assisted Situation Report Generation for Intelligence Analysts

3.4.5.1 Summary of the research item

This research introduces "SmartBook" [12] an AI system designed to assist intelligence analysts in generating situation reports. SmartBook leverages large language models (LLMs) to process vast amounts of data and translate it into concise and informative reports. The system is trained on a dataset of real world reports and evaluated using several metrics.

3.4.5.2 Critical analysis of the research item

SmartBook offers a promising solution for intelligence analysts by automating time-consuming report generation tasks.

3.4.5.3 Relationship to the proposed research work

This research is relevant to our project as it showcases the potential of using large language models (LLMs) for report generation, which aligns with our goal of providing detailed, AI-generated sales and customer interaction reports and stakeholders reports.

3.5 Literature Review Summary Table

The Related Work Summary is shown in Table 3.3

Table 3.3: Summary of Reviewed Literature

Paper Title	**Method**	**Results**	**Limitations**
DialogBench [3]	Evaluation benchmark with 12 dialogue tasks	Comprehensive evaluation of LLMs' human-like capabilities	Subjective assessment of human-likeness, potential biases from GPT-4
AI Coach Assist [9]	Pre-trained Model	Improved training for contact center agents	High computational needs, limited to specific languages
Chat GPT [10]	Integration with voice assistants	Speaking and listening practice with feedback	Larger studies needed for generalizability
WavLM_SI [8]	Model for real-time speech interruption detection	Improved True Positive Rate and reduced model size for deployment	Performance in environments with background noise is unaddressed
SmartBook [12]	LLM-based report generation	Efficient generation of concise situation reports	Generalizability to other report types is uncertain

3.6 Conclusion

The literature review has shed light on key advancements in AI-driven automation for sales and customer care. By evaluating the strengths and weaknesses of various products and relevant research papers, we aim to ensure that our project not only builds on their strengths but also addressing their weaknesses.

Chapter 4 Software Requirement Specifications

In this chapter, we discuss the functionality and expected performance of the AI Automation for Sales and Customer Care system. The system requirements and use cases will also be covered.

4.1 List of Features

The key characteristics of the AI Automation for Sales and Customer Care system include:

- CRM Data Integration and Lead Scoring
- AI-Driven Phone Calls and Product Pitching
- Advanced Reporting and Analytics
- Customer Support Chatbot
- Order Placement During Calls
- Call Feedback and Sentiment Analysis
- Automated Real-Time Report Generation

4.2 Functional Requirements

4.2.1 User Management

- **Sign Up** The system will allow new users to register accounts by providing necessary personal information like their name, email, and password.
- **Login** The system will enable current users to access their accounts with their credentials (email and password) and ensure their session is secure.
- **Profile Management** Users are able to see and modify their profile details, such as name, email address, and contact information.

4.2.2 CRM Data Integration and Lead Scoring

- The system connects with current CRM systems to retrieve data and handle customer information for sales and support functions.
- The system uses lead scoring algorithms on CRM data, providing scores to potential leads according to criteria like customer engagement, history of interactions, and buying patterns.

4.2.3 AI-Driven Phone Calls and Product Pitching

- The system will employ LLMs to make phone calls to customers, utilizing CRM data to provide customized product suggestions and services.
- While on calls, customers will have the ability to place orders, and any feedback they give will be recorded for later analysis.

4.2.4 Customer Support Chatbot

- The system will feature an AI-driven chatbot that addresses customer questions, offers product details, and escalates concerns to human representatives when required.
- The chatbot will be accessible around the clock to help customers with tasks such as troubleshooting and general questions.

4.2.5 Order Placement During Calls

- Customers will have the ability to make orders during phone calls managed by AI.
- The system will gather and manage order information, merging it with sales data and CRM for reporting purposes.

4.2.6 Call Feedback and Sentiment Analysis

- Following every AI-powered call, the system will solicit input from customers.
- Sentiment analysis will be conducted on the feedback to assess customer satisfaction, and findings will be documented for future enhancements.

4.2.7 Automated Report Generation

- The system will produce real-time reports on essential metrics including sales performance, customer satisfaction, call success rates, and order information.
- Sales teams should be able to customize the reports.

4.2.8 Logout

- The system will enable users to safely log out, ending their session and directing them to the login page.

4.3 Quality Attributes

- **Performance** The system must manage large amounts of customer engagements and report creation without a drop in performance.
- **Reliability** The AI algorithms and CRM integration need to guarantee precise data processing and interactions.
- **Scalability** The system must scale effectively to accommodate more users and growing sales data as time progresses.
- **Security** Sensitive customer information must be safeguarded by employing encryption and secure communication methods.

4.4 Non-Functional Requirements

4.4.1 Performance

The system must manage simultaneous interactions and produce reports with low latency. Calls powered by AI should function seamlessly in real-time.

4.4.2 Reliability

The platform should guarantee dependable access to customer information and interaction records, accompanied by backup and recovery systems.

4.4.3 Scalability

The system must efficiently adapt to the growing number of users, accommodated sales calls, and data storage needs.

4.5 Assumptions

In establishing the specifications and crafting the AI Automation for the Sales and Customer Care system, the subsequent assumptions were considered:

- **Internet Connectivity** It is presumed that every user will be able to access a reliable internet connection while engaging with the system.
- **Device Compatibility** Users will mainly access the system via contemporary web browsers on desktop computers or laptops.

- **CRM Data Availability** It is expected that the CRM data is precise, current, and constantly accessible for immediate interactions.
- **User Input Accuracy** Users should provide accurate and valid input while using the system.
- **User Familiarity** It is presumed that users possess a fundamental knowledge of utilizing web-based systems, such as logging in, navigating through interfaces, and understanding reports or dashboards.

4.6 Use Cases

This section outlines all the use cases for the AI Automation system for Sales and Customer Care, specifying how various actors engage with the system.

4.6.1 Use Case 1: Sign Up

Name	Sign Up
Actors	Customer, Admin
Summary	The system allows new users to create an account by providing name, email, and password.
Pre-Conditions	The user must not have an existing account in the system.
Post-Conditions	A new user is created and the user is directed to login page.
Basic Flow	1. The user navigates to the sign-up page. 2. The system prompts the user to enter details like name, email, and password. 3. The system verifies that the email is unique and valid. 4. The user account is created, and the system confirms the registration. 5. The user is redirected to the login page.
Alternative Flow	• 3A. If the email is already registered, the system shows an error message. • 3B. If the user enters invalid information, the system prompts them to correct the input.

4.6.2 Use Case 2: Login

Name	Login
Actors	Customer, Admin
Summary	The system authenticates users and grants access.
Pre-Conditions	The user must have a registered account with the system.
Post-Conditions	The user is successfully authenticated and redirected to dashboard.
Basic Flow	1. The user enters their email and password on the login page. 2. The system verifies the credentials. 3. If valid, the user is logged in and redirected to the dashboard. 4. If invalid, an error message is displayed.
Alternative Flow	• 2A. If the user enters incorrect credentials, the system displays an error message asking them to try again. • 2B. If the user forgets their password, they can select ”Forgot Password” to reset their password via email.

4.6.3 Use Case 3: Profile View

Name	Profile View
Actors	Customer
Summary	The system allows the user to view their information and details.
Pre-Conditions	The user must be logged into the system.
Post-Conditions	The user’s profile information is displayed on the profile page.
Basic Flow	1. The user navigates to the profile page. 2. The system displays the user account information, such as name, email. 3. The user can view their account and update information.
Alternative Flow	• 1A. If the user is not logged in, they are redirected to login page.

4.6.4 Use Case 4: Profile Update

Name	Profile Update
Actors	Customer
Summary	The system allows users to update their personal details such as name, email, password, or contact information.
Pre-Conditions	The user must be logged in and have access to their profile page.
Post-Conditions	The user's updated information is saved in the system.
Basic Flow	1. The user accesses their profile page and selects the update option. 2. The system prompts the user to enter new information. 3. The user submits the updated details.
Alternative Flow	• 3A. If the user enters invalid or incomplete information, the system prompts the user to correct the data before submission.

4.6.5 Use Case 5: Generate Reports

Name	Generate Reports
Actors	Admin, Sales Team
Summary	The system generates and provides detailed reports on sales performance, customer interactions, and other metrics.
Pre-Conditions	The system must have access to sufficient data on sales records.
Post-Conditions	A report is generated and made available for viewing.
Basic Flow	1. The user selects the option to generate a report. 2. The system gathers relevant data from the database. 3. The system generates the report for insights. 4. The user is notified when the report is ready and can view it.
Alternative Flow	• 2A. If there is insufficient data to generate a report, the system displays a message to the user, notifying them that the report cannot be generated.

4.6.6 Use Case 6: CRM Integration

Name	AI-Powered Schema Mapping for CRM Integration
Actors	• System Administrator • Business Client • Data Integration Platform • AI Language Model
Summary	The system automates CRM data mapping using AI, aligning columns based on meaning rather than names. This reduces integration time and errors compared to manual methods.
Pre-Conditions	• Access to CRM dataset • Predefined target schema rules • Configured AI model
Post-Conditions	• Mapped CRM data in standard format • Incompatible fields flagged for review • Stored dataset available for use
Basic Flow	1. System loads AI model and mapping rules. 2. Client provides source dataset. 3. System analyzes and maps columns semantically. 4. Compatible data is transferred; complex fields are merged. 5. Incompatible columns are flagged. 6. Mapped dataset is saved.
Alternative Flow	• **No compatible column:** Marked as ”Nothing Compatible” for review. • **Multiple matches:** System merges data from relevant columns. • **Errors:** System logs issues and halts the process. • **Dataset structure changes:** System adapts mappings.

4.6.7 Use Case 7: Chatbot Interaction

Name	Chatbot Interaction
Actors	Customer
Summary	The chatbot assists customers with queries, product information, and service requests using AI-based interactions.
Pre-Conditions	The customer must have a valid account and be logged into the system.
Post-Conditions	The customer's query is resolved, or they are connected to a human agent if the chatbot cannot handle the request.
Basic Flow	1. The customer initiates a conversation with the chatbot. 2. The chatbot processes the customer's query 3. On successful resolution of query the conversation ends. 4. If the chatbot cannot resolve the query, it is handed to an operator
Alternative Flow	• 2A. If the chatbot fails to understand the query, it asks the customer to rephrase the question.

4.6.8 Use Case 8: Logout

Name	Logout
Actors	Customer, Admin
Summary	The system logs the user out and ends their session, preventing further access until they log in again.
Pre-Conditions	The user must be logged into the system.
Post-Conditions	The user is logged out, and their session is terminated.
Basic Flow	1. The user selects the "Logout" option from the menu. 2. The system confirms the logout request. 3. The system terminates the session.
Alternative Flow	• 2A. If the user's session has already expired, the system redirects the user to the login page.

4.7 Hardware and Software Requirements

This section outlines the necessary hardware and software components required to successfully develop, deploy, and maintain the project. These requirements ensure optimal performance and scalability for the system, supporting its core functionalities and user interactions.

4.7.1 Hardware Requirements

The following hardware is essential for the project's implementation:

- Modern server or cloud infrastructure: Required to process web requests, execute AI algorithms, and manage data efficiently. Cloud providers like AWS, Azure, or Google Cloud can be utilized.
- High-speed internet connection: Crucial for enabling seamless real-time customer interactions and data processing.

4.7.2 Software Requirements

- Programming languages: Python, JavaScript
- Web frameworks: Django, FastAPI, Next.js
- Database: PostgreSQL or MySQL, ChromaDB or Qdrant
- AI Tools: TensorFlow or PyTorch
- Third-party APIs: Llama 3.2 or Mistral, Bland AI

4.8 Graphical User Interface

This section presents the graphical user interface (GUI) designs, showcasing how users will interact with the system. The GUI is designed to be intuitive and user-friendly, ensuring a seamless experience for both technical and non-technical users.

Each figure in this section provides a detailed illustration of the layout, features, and navigation flow of the key screens. The interface focuses on delivering clarity and efficiency, enabling users to access functionalities with minimal effort even if they are from a non-technical background. The following figures provide a comprehensive view of the system's GUI, emphasizing the ease of navigation and minimal design.

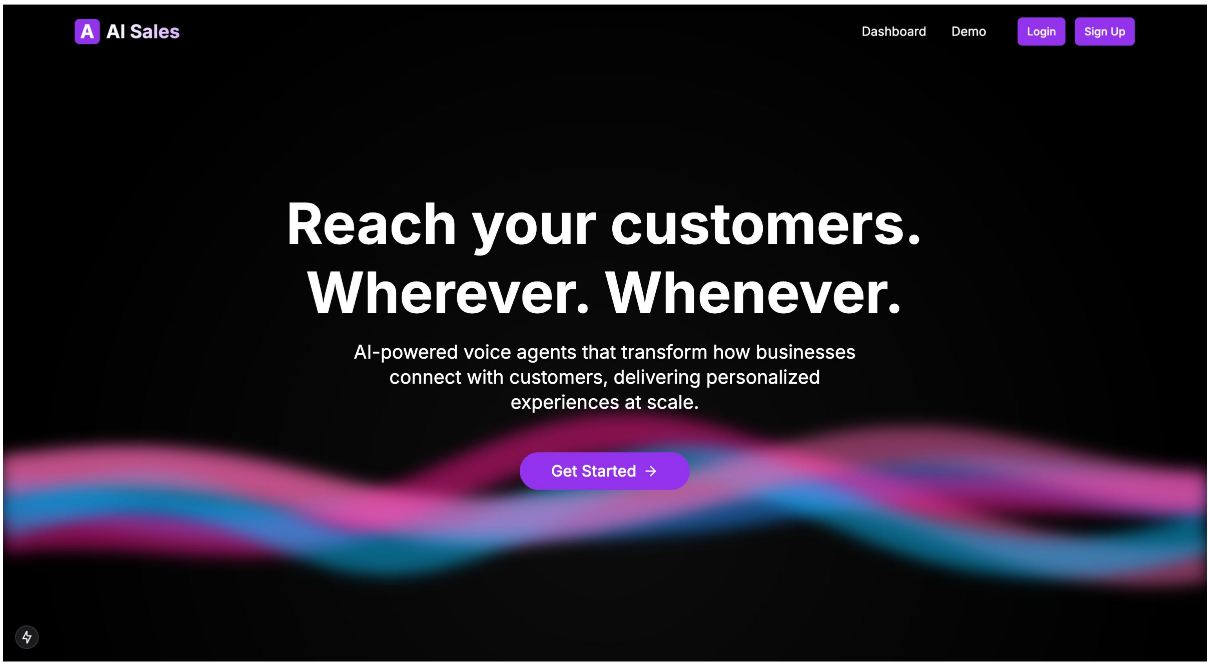

Figure 4.1: Landing Screen

Landing Page for new users

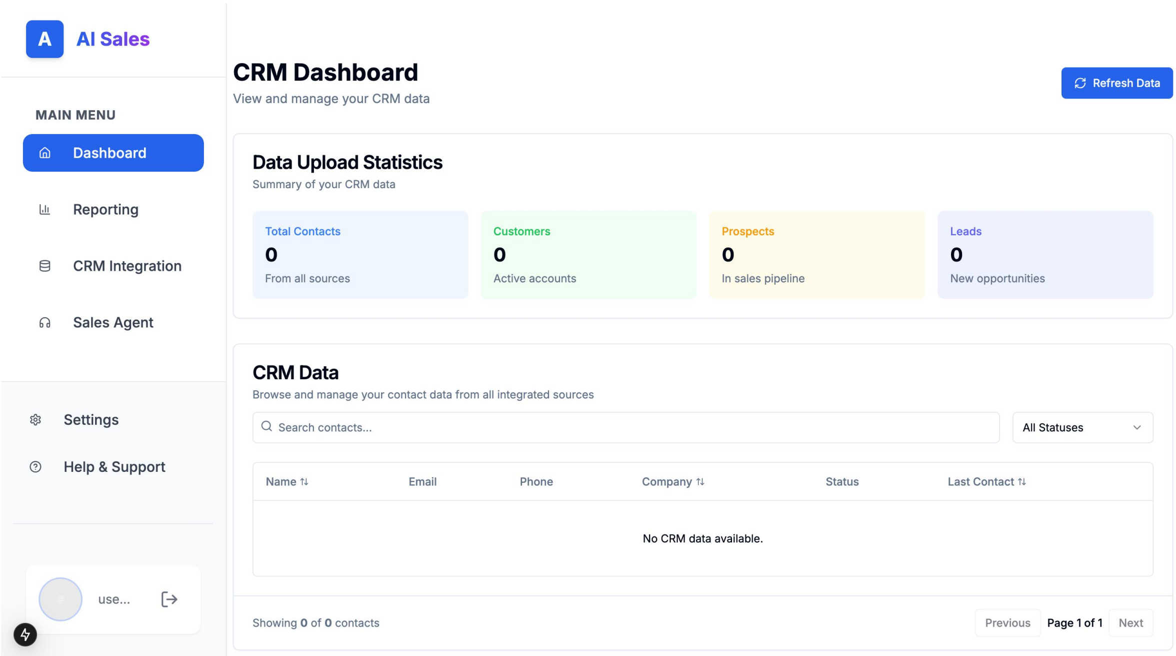

Figure 4.2: Home Screen

Dashboard for the user.

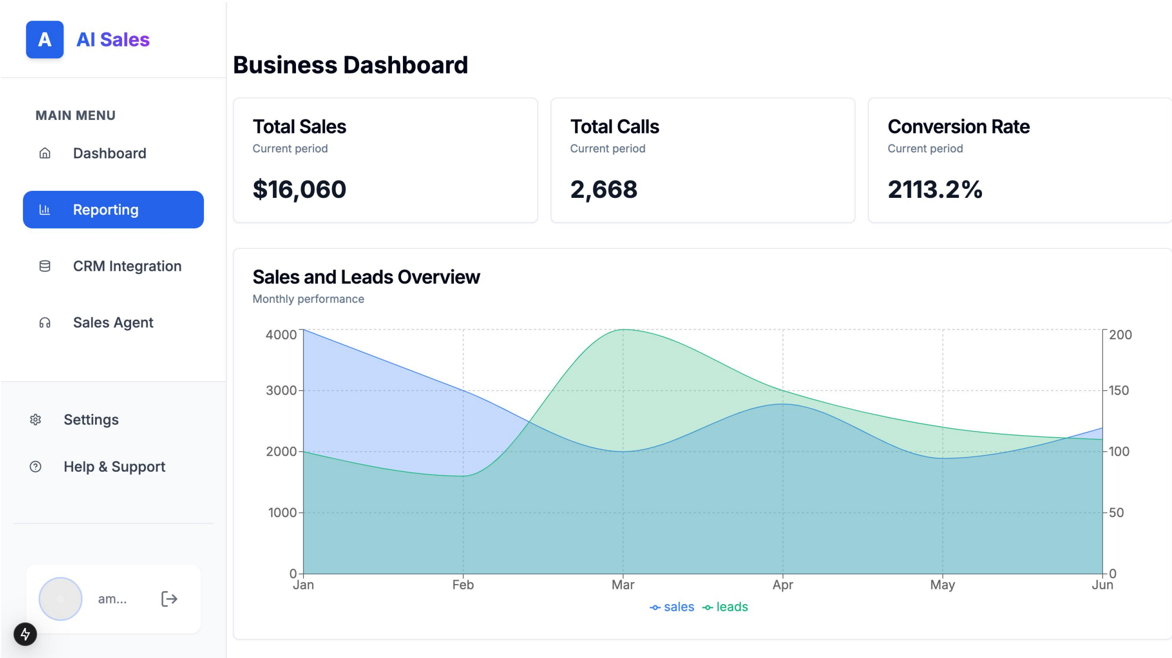

Figure 4.3: Reporting Screen

Reports page for the user.

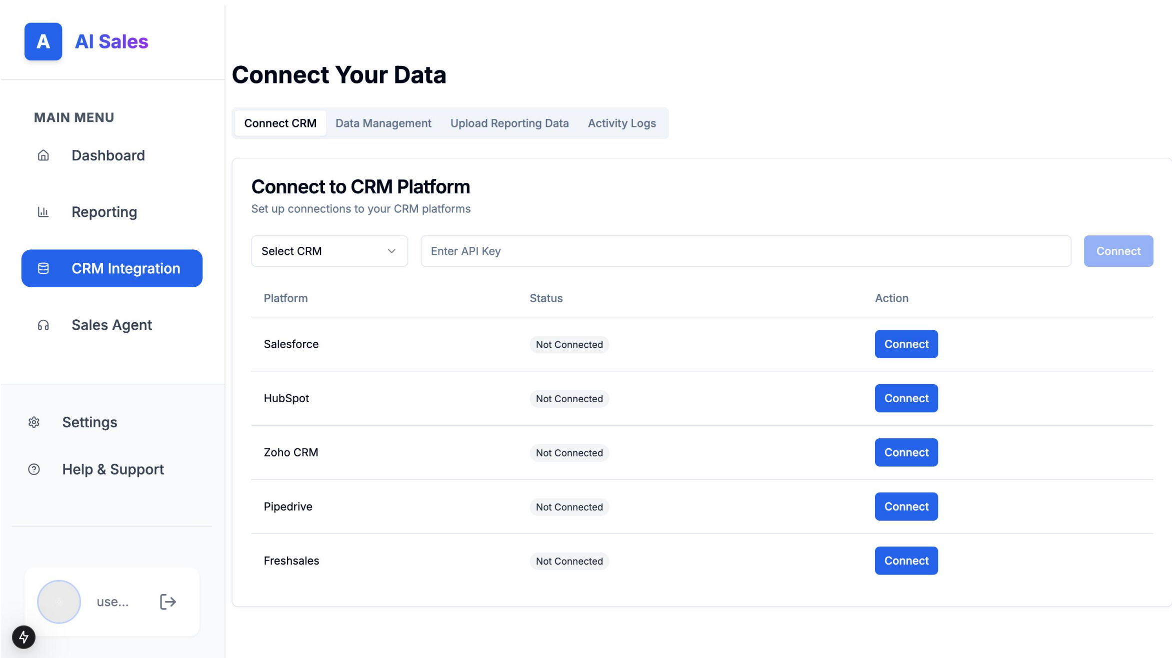

Figure 4.4: CRM Integration Screen

CRM Integration page for the user.

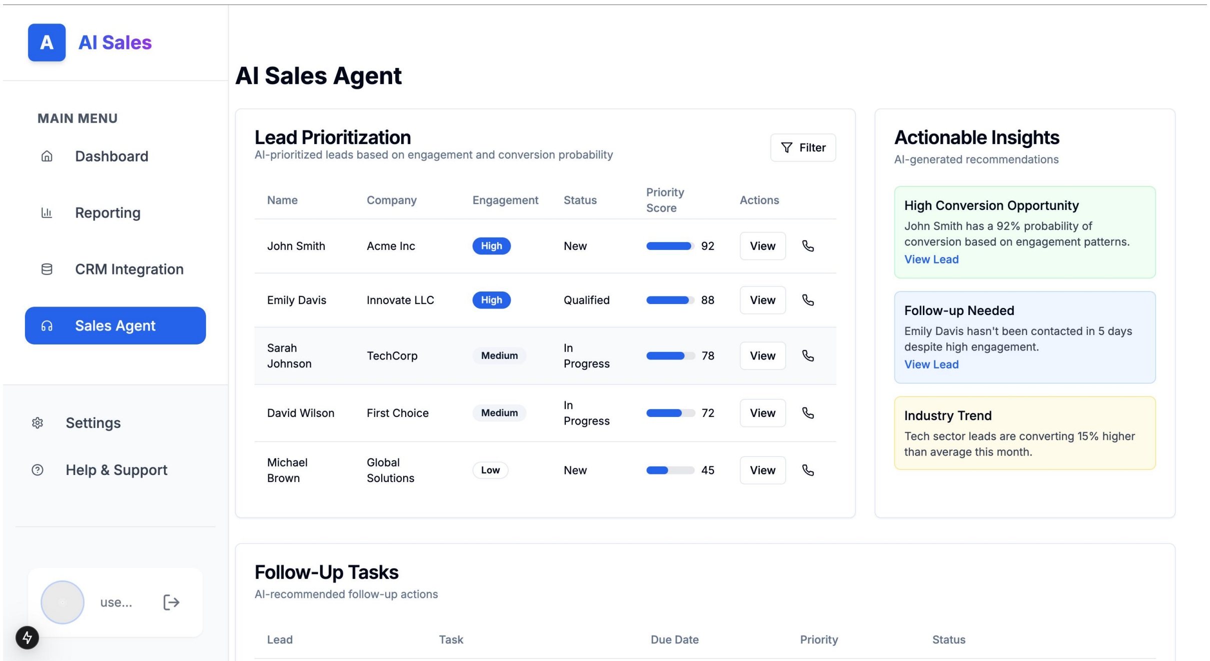

Figure 4.5: Sales Agent Screen

Sales Agent page for the user.

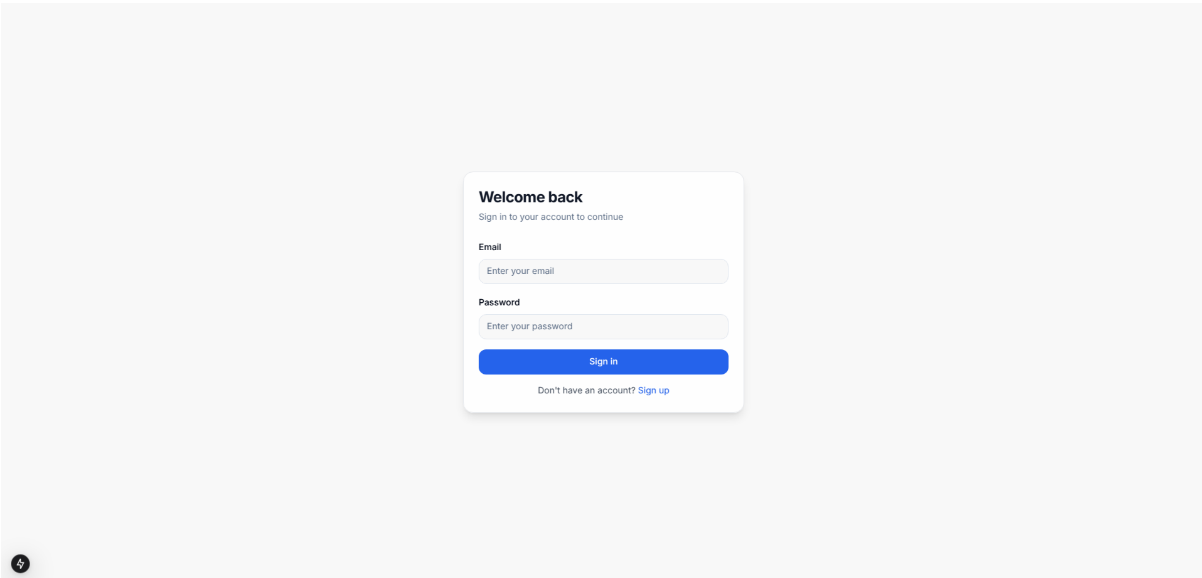

Figure 4.6: Login Screen

Login page for the user.

4.9 ER Diagram

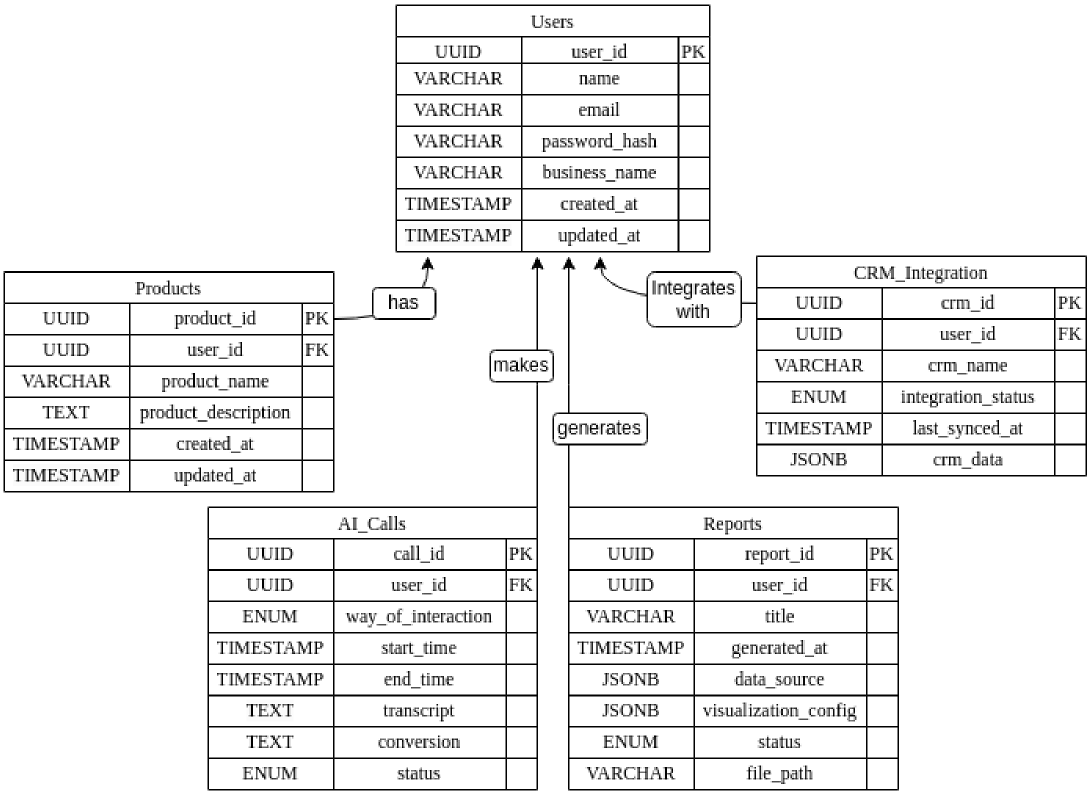

Figure 4.7: Database design

4.10 Data Dictionary

Table 4.1: Data Dictionary - Part 1

Entity	Attribute	Data Type	Nullable	Description
Users	user_id	UUID	No	The unique identifier for a user.
	name	String	No	The user's full name.
	email	String	No	The registered email address of the user.
	password_hash	String	No	The encrypted password for authentication.
	business_name	String	No	The name of the business associated with the user.
	created_at, updated_at	Timestamp	No	Timestamps indicating when the record was created and last updated.

Table 4.2: Data Dictionary - Part 2

Entity	Attribute	Data Type	Nullable	Description
AI Calls	call_id	UUID	No	The unique identifier for an AI interaction (call, message, or email).
	user_id	UUID	No	Foreign key referencing the user who initiated the interaction.
	way_of_interaction	Enum	No	The type of interaction (e.g., Call, Message, Email).
	start_time, end_time	Timestamp	No	Timestamps for when the interaction started and ended.
	transcript	Text	Yes	The AI-generated transcript of the interaction.
	conversion	Text	Yes	Key insights or outcomes extracted from the interaction.
	status	Enum	No	The status of the interaction (e.g., Completed, Failed).
CRM Integration	crm_id	UUID	No	The unique identifier for a CRM integration record.
	user_id	UUID	No	Foreign key referencing the user integrating the CRM.
	crm_name	String	No	The name of the CRM system (e.g., Salesforce).
	integration_status	Enum	No	The status of the integration (e.g., Active, Inactive).
	crm_data	JSONB	Yes	Metadata or serialized data fetched from the CRM system.

Table 4.3: Data Dictionary - Part 3

Entity	Attribute	Data Type	Nullable	Description
Products	product_id	UUID	No	The unique identifier for a product.
	user_id	UUID	No	Foreign key referencing the user who owns the product.
	product_name	String	No	The name of the product.
	product_description	Text	Yes	A detailed description of the product.
	created_at, updated_at	Timestamp	No	Timestamps indicating when the record was created and last updated.
Reports	report_id	UUID	No	The unique identifier for a report.
	user_id	UUID	No	Foreign key referencing the user who generated the report.
	title	String	No	The title of the report.
	generated_at	Timestamp	No	Timestamp when the report was generated.
	data_source	JSONB	Yes	Metadata about the report's data source(s).
	visualization_config	JSONB	Yes	Configuration for report visualizations (e.g., chart types).

4.11 Risk Analysis

- **AI Model Accuracy** Inaccuracies in customer interaction could lead to unsatisfactory experiences.
- **Integration Challenges** Difficulties in integrating with various CRM systems.
- **Data Privacy** Handling sensitive customer information requires robust security protocols.

List and explain the risks that may be encountered during the project. For e.g.: technical risks, business risks, etc.

Chapter 5 High-Level and Low-Level Design

This chapter discusses the high-level and the low-level design for AI Automation for Sales and Customer Care. The high-level design focuses on the system architecture and major components while the low-level design is a deeper dive into each of the component. The low-level design will focus more on the in depth details and the specific parts of each module, the classes, methods. Both the high-level and low-level designs are important steps in building AI Automation for Sales and Customer Care.

5.1 System Overview

The project is designed to revolutionise how businesses interact with clients. The system integrates with existing CRM systems to create comprehensive sales and customer service operations.

5.1.1 Core Functionality

- **CRM Integration** The system can integrate with existing CRM systems using APIs. Most CRM systems, like Salesforce, HubSpot, Zoho, and Microsoft Dynamics, offer RESTful APIs that allow external platforms to interact with the CRM's data. We can use OAUTH2 or API keys to authenticate the app with the CRM system. However, the user will be given the option to simply upload the CRM data to the app.
- **AI-Driven Voice Interactions** We will use existing AI audio models to develop multi-agents that can initiate and manage phone calls with customers. These interactions are designed to engage customers, pitch products, and gather valuable feedback.
- **Intelligent Chatbot** We will utilize existing LLMs such as Llama3.2 for analyzing and Gemma2 for chatting. The chatbot will be capable of handling complex queries in English, offering solutions and product information, and routing to human agents when necessary, as multi-agent models will be developed.
- **Automated Reporting** The system generates comprehensive reports on sales performance, customer satisfaction, and ROI. These reports include visualizations and can be produced on a monthly or annual basis.
- **Call Analysis and Feedback Processing** Using speech-to-text technology and sentiment analysis, the system processes recorded calls to generate detailed reports on customer interactions.
- **Semi-Automated Order Placement** While not directly placing orders, the system creates trigger events for potential sales, involving human oversight in the final ordering process and sending

confirmation emails to customers.

5.1.2 System Architecture

- **Integration Layer** Handles connections with various CRM systems and other external services.
- **AI Core** Manages voice AI, natural language processing, and machine learning models.
- **Interaction Management** Coordinates customer interactions across voice and chat channels.
- **Analytics Engine** Processes data for reporting and feedback analysis.
- **User Interface** Provides dashboards for system management and report visualization.

5.2 Design Considerations

5.2.1 Assumptions and Dependencies

5.2.1.1 Assumptions

- **CRM System Usage** The system assumes that target businesses use modern CRM systems with API access. Additionally, users will be given the option to upload CRM data manually, but this comes with performance risks.
- **End-User Proficiency** End-users (business staff) have basic technical proficiency to interact with the system's interface efficiently.
- **Legal Compliance** The use of AI for customer interactions complies with current legal standards in the jurisdictions where the platform is used.
- **Hardware Availability** It is assumed that the businesses using the platform have sufficient hardware resources, such as servers or cloud infrastructure (AWS, AZURE, HuggingFace Inference), to support AI processing tasks like voice interactions and report generation.
- **Operating System Compatibility** The platform will be deployed in environments running widely supported operating systems such as Linux, macOS, or Windows.
- **Future AI Upgrades** The system will need to support upgrades to the AI models (e.g., LLaMA) and may require the addition of more advanced features, such as multi-language support or voice cloning.

5.2.1.2 Dependencies

- **CRM Data Quality and Accessibility** The effectiveness of the system heavily depends on the quality and accessibility of data in the integrated CRM systems. Inaccurate or incomplete data may reduce the system's effectiveness.
- **User Technical Skills** User adoption and system effectiveness may vary based on the technical proficiency of the staff using it. Staff with limited technical skills may require additional training or support.
- **Legal Changes Regarding AI** Changes in laws and regulations regarding the use of AI in customer service may impact the system's functionality, requiring adjustments or updates to ensure compliance.
- **Hardware Performance Requirements** For optimal performance, the system will require access to modern hardware, such as GPUs, for running AI models (e.g., LLaMA for speech-to-text or sentiment analysis).
- **Compatibility with Libraries and Operating Systems** The AI models (such as LLaMA) and supporting libraries (like LangChain) are dependent on compatible operating systems. Updates to these libraries or operating systems may necessitate code changes or library upgrades to maintain functionality.
- **Business Requirement Changes** Any changes in business requirements, such as the need to support additional CRM systems or new data privacy regulations, could drive changes in the platform's core functionality, requiring further development or architectural adjustments.

5.2.2 General Constraints

1. **Hardware or Software Environment** The system is designed to run on modern hardware capable of supporting AI models (e.g., GPUs for LLMs and other machine learning models). Businesses without access to this hardware may experience performance lag. Limited hardware resources may affect the real-time capabilities of AI-driven functionalities.
2. **Availability or Volatility of Resources** The availability of customer data from integrated CRM systems is critical to the system's operation. If the CRM system is down or the API access is restricted, core functionalities like data-driven AI calls and reporting will be impacted. The system's reliance on third-party CRM data increases vulnerability to external downtimes, API changes, or data inaccessibility, which may cause service disruptions.
3. **Interoperability Requirements** The system must support integration with multiple CRM sys-

tems, requiring flexible and modular API handling. The need for interoperability may complicate the design, as the system will have to accommodate various data structures, APIs, and CRM workflows.

4. **Verification and Validation Requirements (Testing)** Continuous verification and validation processes are needed to ensure the accuracy of AI-driven actions (e.g., speech-to-text conversion, product pitching). Inadequate testing may lead to incorrect data processing, AI errors, or inaccurate reporting, which could harm the user experience and reduce the platform's reliability.

5. **Resource Allocation** Each agent requires access to computational resources (CPU, memory, and potentially GPU) to function efficiently, especially for tasks like speech recognition, natural language understanding, and report generation. Limited resources could lead to slow agent responses, degraded system performance, or agents becoming unresponsive. Resource management techniques like dynamic allocation or load balancing are necessary to prevent resource exhaustion.

6. **Inter-Agent Dependencies** Some agents may depend on the output or actions of others (e.g., the reporting agent may depend on the data collected by the AI call agent). Failure in one agent (e.g., incomplete data collection) could cascade to other agents, resulting in incomplete reports or incorrect recommendations. Clear dependencies and fallback mechanisms are needed to ensure system resilience.

7. **Performance Requirements** If performance is compromised, it could affect customer engagement and reduce the overall effectiveness of the system, making the automation less attractive to businesses.

8. **Other Quality Goals** The system must maintain high availability and reliability, ensuring that critical components like AI calls and CRM integration work seamlessly. Achieving these quality goals requires a robust infrastructure and frequent updates to handle potential errors or downtimes effectively.

9. **Network Communications** The system relies on stable and fast network communications. Any network disruptions or slow internet speeds can significantly affect the platform's functionality, leading to delays in communication and processing.

5.2.3 Goals and Guidelines

- **Simplicity and Maintainability (KISS Principle)** Simple and maintainable modular design will be followed to reduce development complexity and allow for easier updates, ensuring long-term sustainability.

- **Performance and Scalability** Our system will prioritize fast and scalable AI-driven interaction as real-time customer engagement and reporting are critical tasks and require fast computing and up-to-date hardware.
- **Modular Design** For future upgradeability , our system will follow a modular approach.
- **Security** It will be ensured that our system is prepared to handle sensitive customer data.
- **Efficient Resource Utilization** The system will be optimised for hardware requirements to reduce cost.
- **CRM Integration** Ensure smooth integration with various CRM platforms as Real-time CRM data access is crucial for AI-powered interactions and accurate reporting.

5.2.4 Development Methods

Scrum methodology will be used for development. Scrum is an iterative and incremental agile framework and is used for continuous improvement throughout the project lifecycle. This methodology is perfect for our system and team , allowing the team to adapt quickly to changes and integrate new features or requirements. The period for each sprint will be at max one week (tentative in case of exceptions). After the completion, we will have a short meeting. In these meetings, we will discuss the work, review the progress, and schedule the rest of the work

5.3 System Architecture

- **CRM Integration and Admin Configuration** The Admin configures and integrates the CRM system. This System syncs data with the CRM Database.
- **Client Classification Process** A Scheduled Task triggers the Client Classification Service. This service fetches data from the CRM Database and classifies potential clients. The result is a Potential Client List.
- **AI Driven Interaction** The AI Interaction Scheduler selects clients from the Potential Client List. It then schedules interactions through various channels (Voice, Text, Email).
- **Multichannel Client Interaction** The system initiates contact with customers through Voice AI, Chatbot, or Email services. .
- **Core Services and Processing** Customer interactions are routed through the API Gateway to Core Services. These services will analyze calls, manage orders, and generate reports.
- **Human in the Loop** Human agents can assist or take over interactions when necessary.

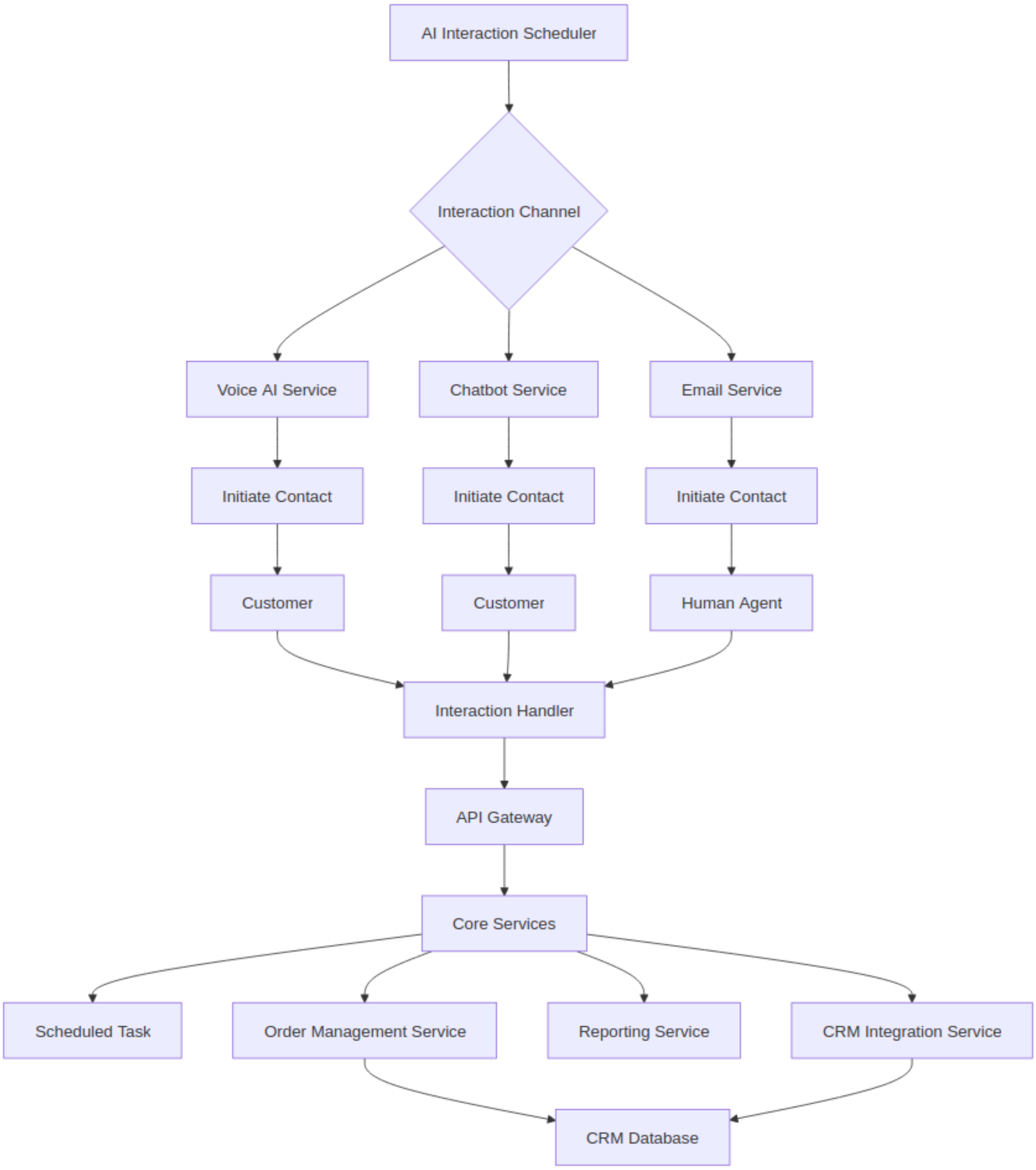

Figure 5.1: High level design of the Project

5.3.1 Subsystem Architecture

5.3.1.1 Call Analytics Subsystem

The Call Analytics subsystem is responsible for processing and analyzing all customer interactions, including voice calls, chat sessions, and email communications. It employs natural language processing and machine learning algorithms to extract meaningful insights from these interactions. The subsystem consists of a data ingestion module that captures raw interaction data, a processing engine that applies various analytical models, and a results aggregator that compiles the findings. It interfaces with the CRM Integration service to correlate interaction data with customer profiles and history. The analytics engine is designed to be extensible, allowing for the easy addition of new analytical models as requirements evolve. Results from this subsystem feed into the Reporting service for stakeholder insights and the Client Selection subsystem to refine targeting criteria.

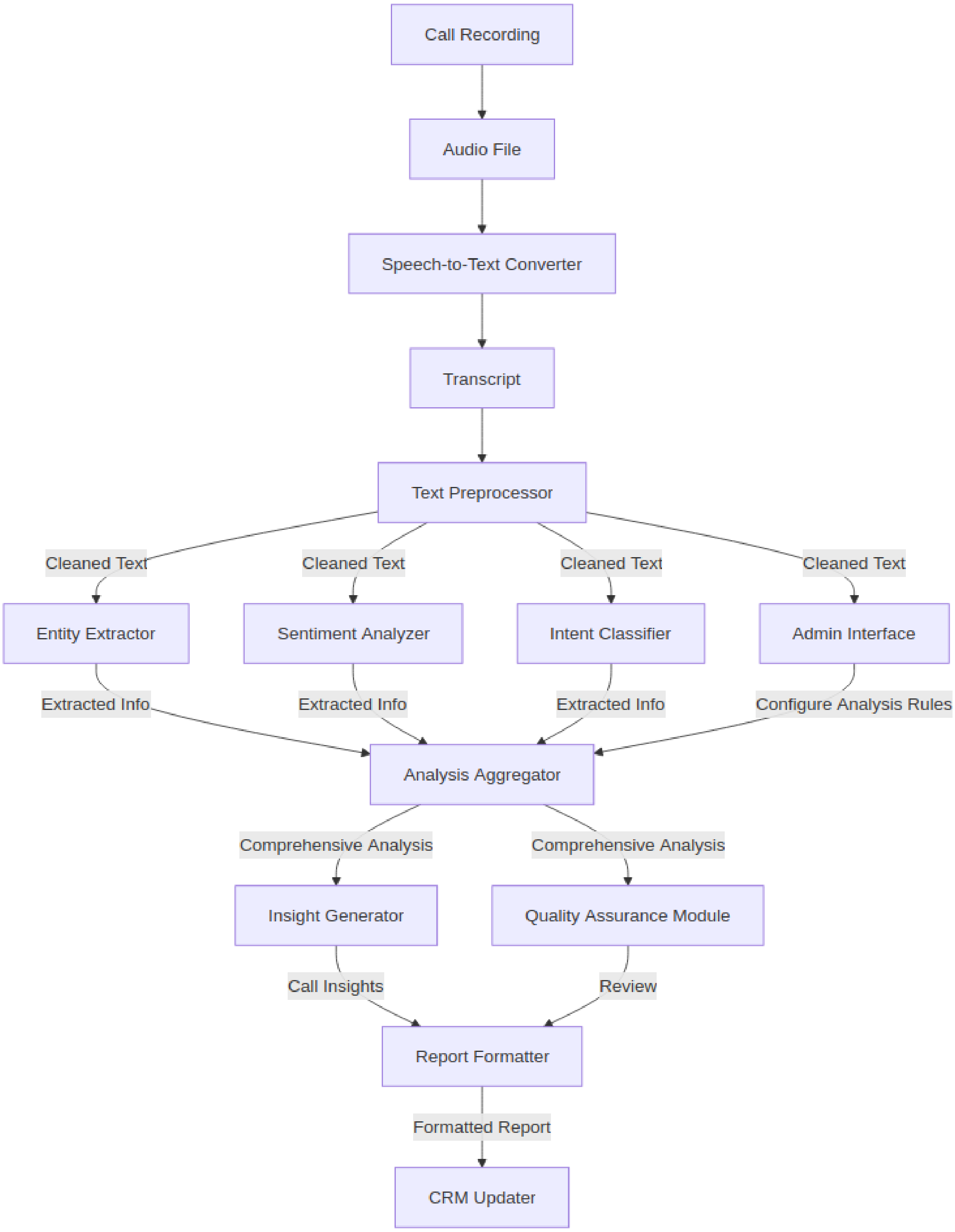

Figure 5.2: Architecture Management and Analysis

5.3.1.2 AI Call Subsystem

The AI Call subsystem manages all aspects of AI-driven voice interactions with customers. It comprises a voice synthesis module for natural-sounding speech output, a speech recognition component for understanding customer responses, and a conversation management engine that guides the flow of the call based on predefined scripts and real-time analysis of customer responses.

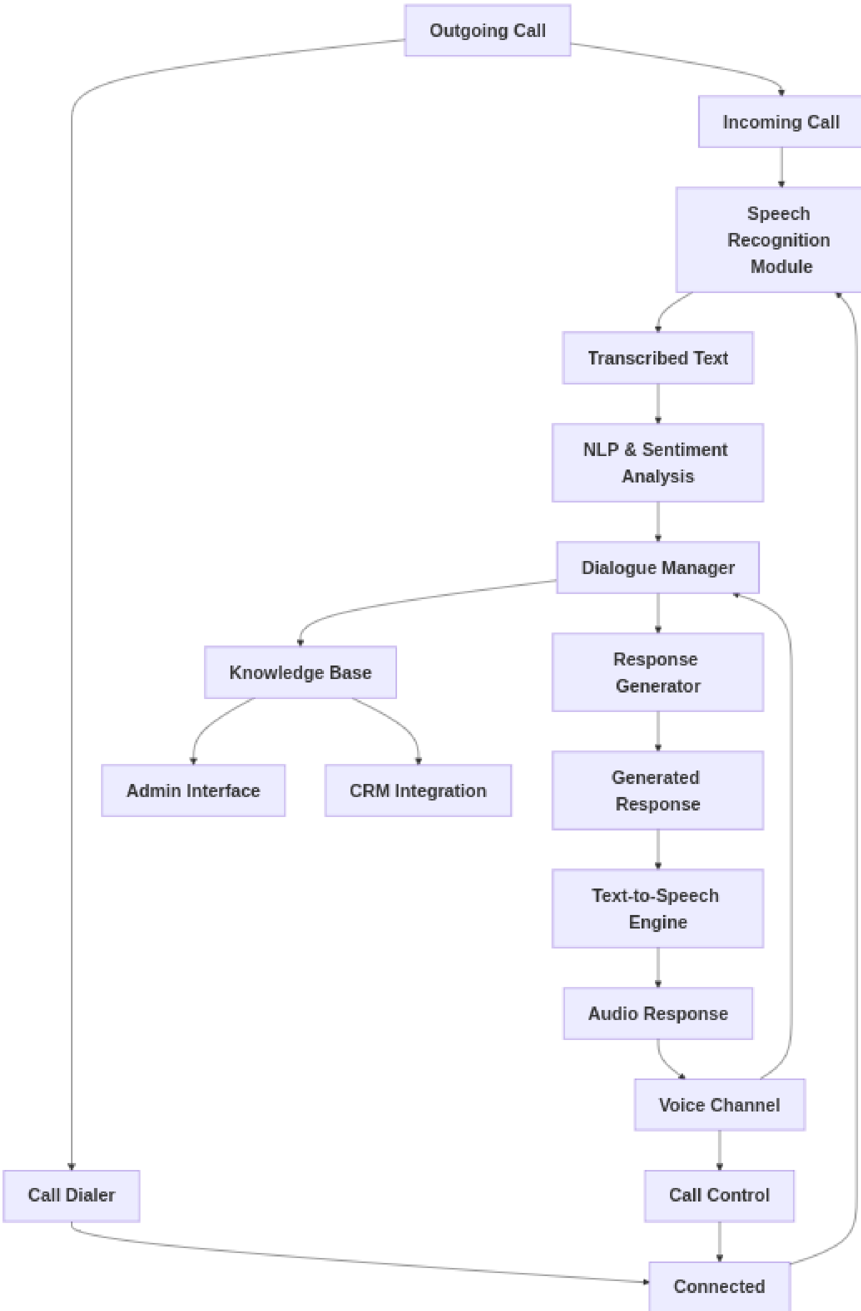

Figure 5.3: AI calls workflow Architecture

5.3.1.3 Client Selection Subsystem

The Client Selection subsystem is tasked with identifying and prioritizing potential clients for outreach from the pool of contacts in the CRM system. It employs a multi-faceted scoring algorithm that considers factors such as past interaction history, purchase patterns, demographic data, and behavioral indicators. The subsystem includes a data enrichment module that augments CRM data with information from external sources to improve selection accuracy.

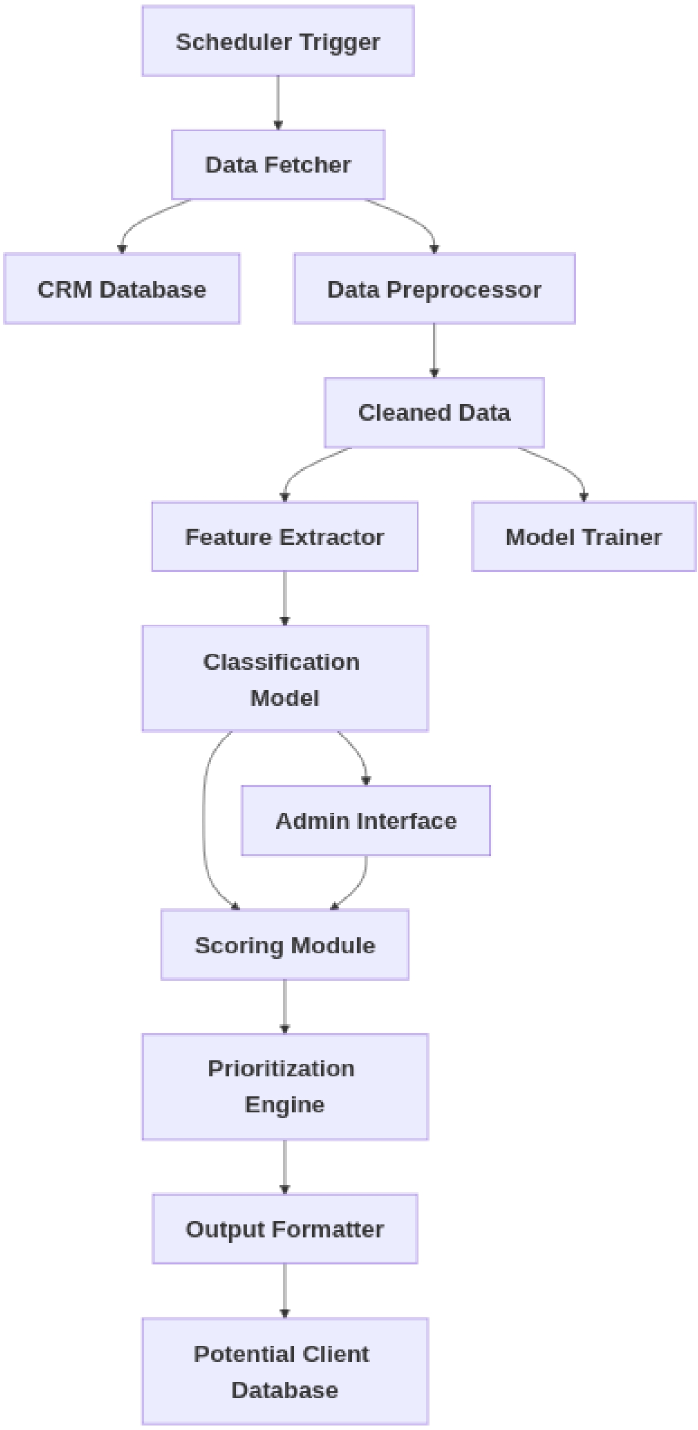

Figure 5.4: Architecture of Client Selection Subsystem

5.3.1.4 Scheduler Subsystem

The Scheduler subsystem orchestrates the timing and channel selection for all automated customer interactions. It consists of a central scheduling engine that manages a queue of pending interactions, a channel optimization module that determines the best medium (voice, chat, email) for each interaction based on customer preferences and historical engagement data, and a workload balancer that distributes interactions across available AI and human resources. The subsystem implements a priority queueing mechanism to ensure time-sensitive interactions are handled promptly. It interfaces with the Client Selection subsystem using microservices to receive lists of targeted clients, the AI Call and Chatbot services to initiate interactions, and the CRM Integration service to update interaction outcomes. The Scheduler also includes a conflict resolution component to prevent multiple simultaneous interactions with the same potential customer.At the end the Task Generator can initiate a call , do chat with the customer or provide email service.

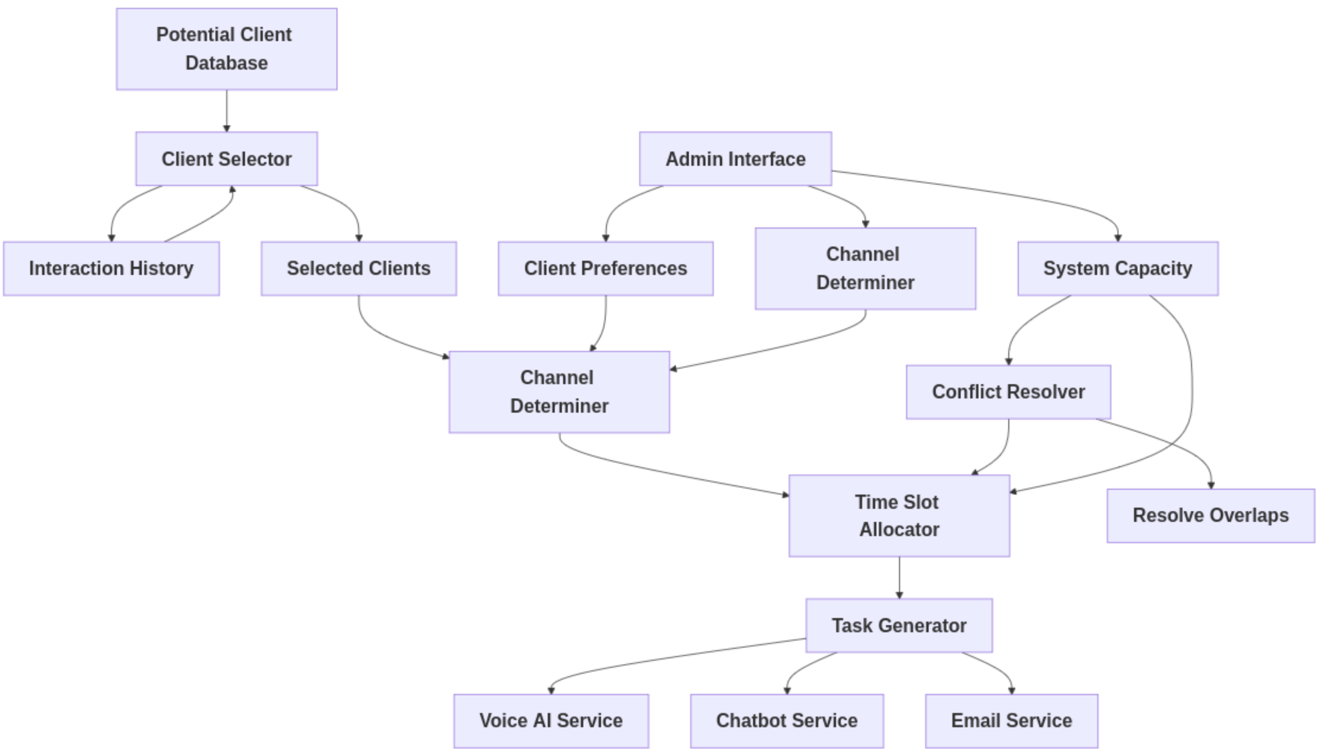

Figure 5.5: AI Scheduler Subsystem

5.4 Architectural Strategies

5.4.1 Tech Stack

For the frontend, we will be using NEXT.js and for the backend, we will use FAST-API. MongoDB will be used as the primary database, and for the vector database, we will use ChromaDB. For reporting and calling, we will use multiple LLMs depending on their capabilities, such as Phi-5 or LLaMA-3.2. For handling multiple AI agents and RAG components, we will use LangChain and LlamaIndex.

5.4.2 User Interface Paradigm

In our sales agent project, we will implement Schneiderman's Eight Golden Rules for the user interface design. These principles will guide the layout, flow, and interactivity, ensuring that the system is intuitive and user-friendly. By adhering to these rules, we aim to streamline the user experience, making it simple and efficient for users to interact with the platform. This approach will help us create an interface that feels natural and minimizes unnecessary complexity, ultimately boosting productivity and satisfaction.

5.4.3 Error Detection and Recovery

Ensuring the reliability of the system involves a comprehensive approach to error detection and handling across both the front end and back end. These logs serve as a valuable tool for identifying the root causes of issues, which can then be addressed to improve the system's performance and stability over time. Such proactive error management contributes to a more reliable and user-friendly experience, fostering trust in the system's operation and performance.

5.4.4 Hosting and Version Control

GitHub will be used for hosting our repository, and for version control, we will be using Git. Version control is important as it will keep the team synchronized and updated. For development, the frontend will be hosted on Vercel, and the backend will be deployed locally.

5.4.5 Reuse of Existing Software Components

For our project, we will utilize existing large language models (LLMs) such as Gemma2 and Llama2, along with other machine learning models. Our main objective is to build on top of these pre-trained AI models, leveraging their capabilities to enhance and tailor them for our specific use case. This approach allows us to save time and resources by utilizing the powerful features of these models, while also focusing on optimizing their performance for the sales agent system.

5.4.6 Concurrency and Synchronization

Asynchronous operations will be implemented for handling backend . By tasks such as data retrieval, processing, and communication with external systems asynchronously, the system ensures that the front-end remains responsive and efficient. This method not only optimizes performance but also contributes to a smoother, more intuitive interaction for the user, ensuring that they can continue with their tasks without unnecessary waiting times.

5.5 Domain Model/Class Diagram

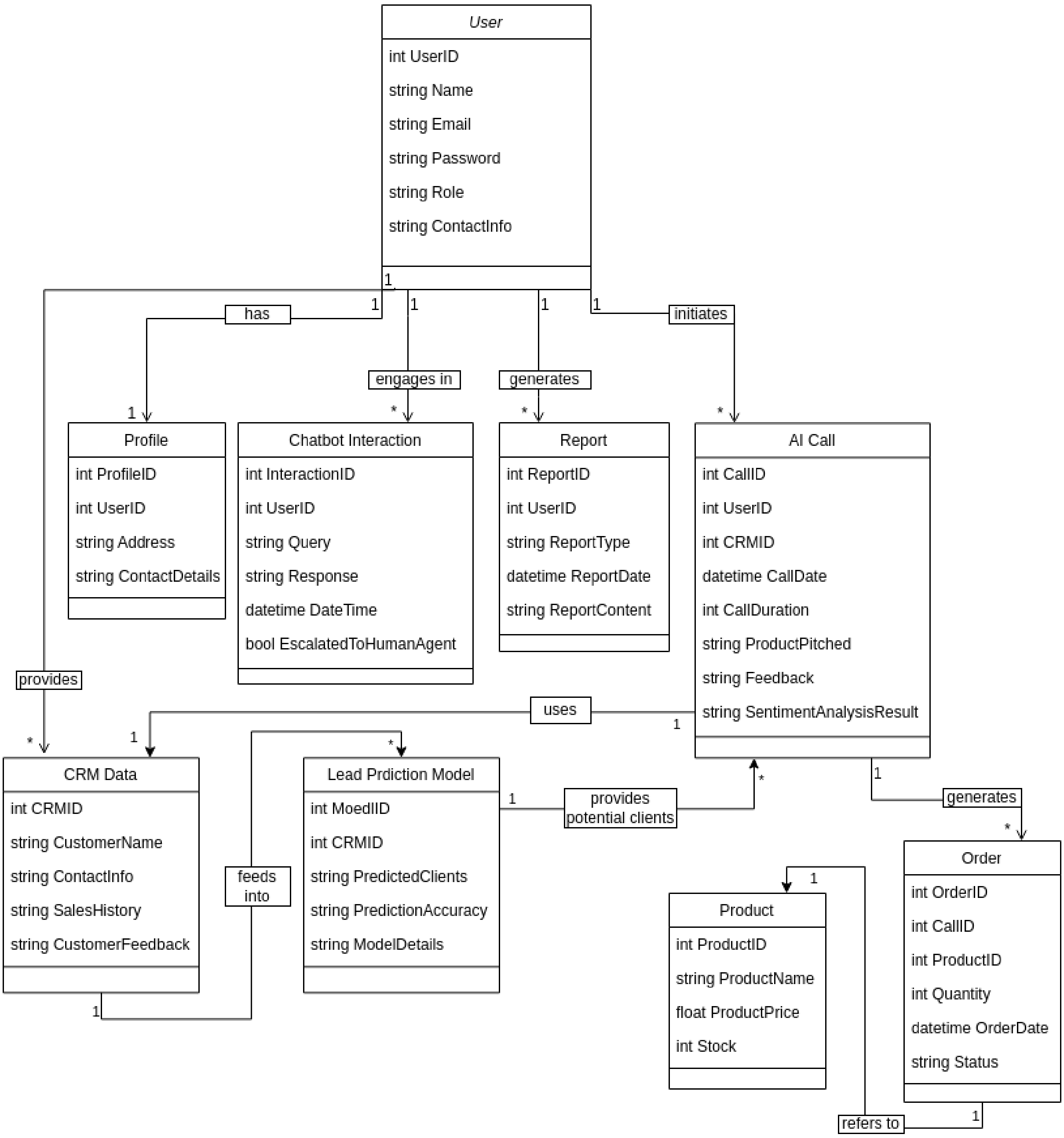

Figure 5.6: Class Diagram

5.6 Policies and Tactics

5.6.1 Microservices Communication Protocol

We have chosen REST as the primary communication protocol between microservices. REST is widely adopted, easy to implement, and fits our needs for simplicity and scalability in communication between services. Additionally, REST is more flexible than gRPC for the variety of clients and platforms we will support. gRPC offers better performance and built-in streaming support, but it was rejected due to the added complexity and our decision to prioritize simplicity; gRPC only performs where we have complex databases.

5.6.2 Data Persistence Strategy

We will implement a polyglot persistence strategy, using different databases tailored to the specific needs of each service. This will optimize performance and scalability across the system.

- **CRM Integration Service** PostgreSQL will be used for structured customer data.
- **Interaction Storage** MongoDB will be used to store varied interaction data due to its flexible schema.
- **AI Handling** ChromaDB will be used to handle and manage the context and workings of the AI models used.

5.6.3 Open Source AI Models

Our AI models, including those for voice AI, chatbots, and analysis, will be deployed using LangChain and LlamaIndex, with ChromaDB as the vector database. LangChain provides flexible and efficient handling of the AI pipeline, while LlamaIndex offers optimal model deployment capabilities. ChromaDB will be used for vector storage, and MongoDB will serve as the main database for non-vectorized data.

5.6.4 Coding Guidelines and Conventions

We will adhere to Google Style Guides for each programming language used in the project to maintain consistency across the codebase.

5.6.5 Testing Strategy

We will implement a testing strategy that includes unit, integration, end-to-end, and performance tests to ensure the reliability of the system.

- **Unit Tests** To validate individual components.
- **Integration Tests** To verify interactions between services.
- **End-to-End Tests** To simulate complete user journeys.
- **Performance Tests** To measure system responsiveness under load.

5.6.6 Monitoring and Logging

We plan to set up a centralized system for logging and monitoring using the ELK stack (Elasticsearch, Logstash, Kibana).

Chapter 6 Description of Prototype Developed

6.1 Implementation

A Sales Agent representing Company and writing scripts for each potential customer and giving responses and sending to calls management microservice

6.1.1 Implementation of Real-Time Calls

The Real-Time Calls component is designed to ensure seamless and interactive communication.

Calls Management

Twilio is used for handling calls. This includes setting up, routing, and managing calls efficiently, ensuring a smooth experience for both agents and customers.

Speech-to-Text and Text-to-Speech Transcription

- **Whisper Model** Integrated via the Grok API, this model transcribes audio from live calls into text with high accuracy.
- **ElevenLabs** Used to generate humane text-to-speech and speech-to-text outputs.

Architecture

The Real-Time Calls operates as a microservice and takes responses from the Sales Agent Microservice. FastAPI is used as the backend.

Deployment

This service is hosted on the Twilio platform, taking full advantage of Twilio's robust telecommunication infrastructure for optimized performance and reliability.

6.1.2 Sales Agent

The Sales Agent is using Python as the programming language.

LLMs Used

- **Llama3.1:8B Q4** and **Llama3.2:Q4** are the primary models for now utilized.
- **Ollama** is used for running models locally, **Grok API** is used for hosted LLMs.

State Management

The Sales Agent has the following states as of now:

- **Greeting** Initiates interaction with the customer.
- **Pitching Agent** Presents the product or service to the customer.
- **Closing** Closes the conversation.
- **Q&A Session** Addresses customer queries.

These states are managed using **LangGraph**, which is used for handling multiple LLM states.

Backend and Microservice Architecture

- The backend is implemented using **FastAPI**. The Sales Agent operates as a microservice and interacts with a real-time call microservice that will be deployed on **Twilio**.
- Company data is provided, including scripts from the company and customer information, that the LLM will use to have a conversation with the customer.

Chapter 7 Conclusions

7.1 Conclusion

The ”AI Automation for Sales and Customer Care” project is a transformative leap in how businesses handle customer interactions and sales processes. Addressing critical inefficiencies in manual workflows, the system offers a fully integrated AI-driven solution that automates customer support, sales outreach, and real-time analytics. The project’s objectives include leveraging advanced AI tools such as chatbots, voice assistants, and sentiment analysis to enhance customer satisfaction and streamline operations. It provides an all-round framework for improving the efficiency of business while at the same time maintaining scalability and security, with functionalities such as AI-driven phone calls, CRM integration, and report generation.

Key differentiators of this project include its focus on real-time voice interactions and comprehensive call feedback mechanisms, areas where the existing platforms such as Salesforce Einstein and Zendesk lack. It also addresses some of the limitations of the platforms such as HubSpot in integrating voice-based outreach and reporting tools. These features make sure that the system not only meets but exceeds expectations in modern businesses looking to optimize their workflows. Also, the microservices architecture makes it possible to scale down the platform and support large-size operations with minimal latency and smooth data streaming.

Going by operational efficiency, the system may reach beyond that. There is a reduction in intervention needs and actionable insights as delivered through real-time analytics, empowering businesses to be able to make data-driven decisions. Overall, it sets up the project to change sales and customer care dramatically through the combination of all AI technologies that automate an end-to-end process. This gives business a holistic product.

7.2 Plans for Future Work

Moving forward into FYP 2, the system will focus on further refining CRM integration and enhancing AI-driven features. This will include improving the platform’s ability to handle complex customer queries and streamline reporting to provide businesses with actionable insights. The AI models will also be fine-tuned for greater accuracy and efficiency, ensuring that the platform offers even more personalized customer interactions. Strengthening the backend infrastructure to support scalability and higher data volumes will also be a priority. Advanced reporting features, such as sentiment analysis and detailed performance metrics, will help businesses make data-driven decisions, which will eventually lead to customer satisfaction and better conversion rates. This will enable the platform to fully automate sales and customer care processes, providing an all-inclusive solution to support business growth and optimization.

Bibliography

[1] Ada.ai: Automating customer service interactions. Available:https://ada.ai Accessed: October 6, 2024.

[2] Bland.ai: Ai platform for voice generation. Available:https://bland.ai Accessed: October 6, 2024.

[3] Dialogbench. Available: arXiv:2311.01677v2 Accessed: October 6, 2024.

[4] Hubspot: Comprehensive marketing, sales, and customer service platform. Available: https://hubspot.com Accessed: October 6, 2024.

[5] Salesforce einstein: Enhancing crm capabilities with ai. Available:https://salesforce.com/einstein Accessed: October 6, 2024.

[6] Yellow ai. Available:https://yellow.ai Accessed: October 6, 2024.

[7] Zendesk: Leading customer service platform. Available: https://zendesk.com Accessed: October 6, 2024.

[8] Real-time speech interruption analysis: From cloud to client deployment. Available:https://arxiv.org/abs/2210.13334, 2022. Accessed: October 6, 2024.

[9] Ai coach assist: An automated approach for call recommendation in contact centers for agent coaching. Available:https://arxiv.org/abs/2305.17619, 2023. Accessed: October 6, 2024.

[10] Chat gpt integrated with voice assistant as learning oral chat-based constructive communication to improve communicative competence for efl learners. Available:https://arxiv.org/abs/2311.00718, 2023. Accessed: October 6, 2024.

[11] Outreach.io: Sales engagement platform. Available:https://outreach.io, 2024. Accessed: October 6, 2024.

[12] Smartbook: Ai-assisted situation report generation for intelligence analysts. Available:https://arxiv.org/abs/2303.14337, 2024. Accessed: October 6, 2024.